

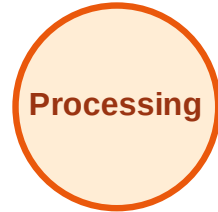
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



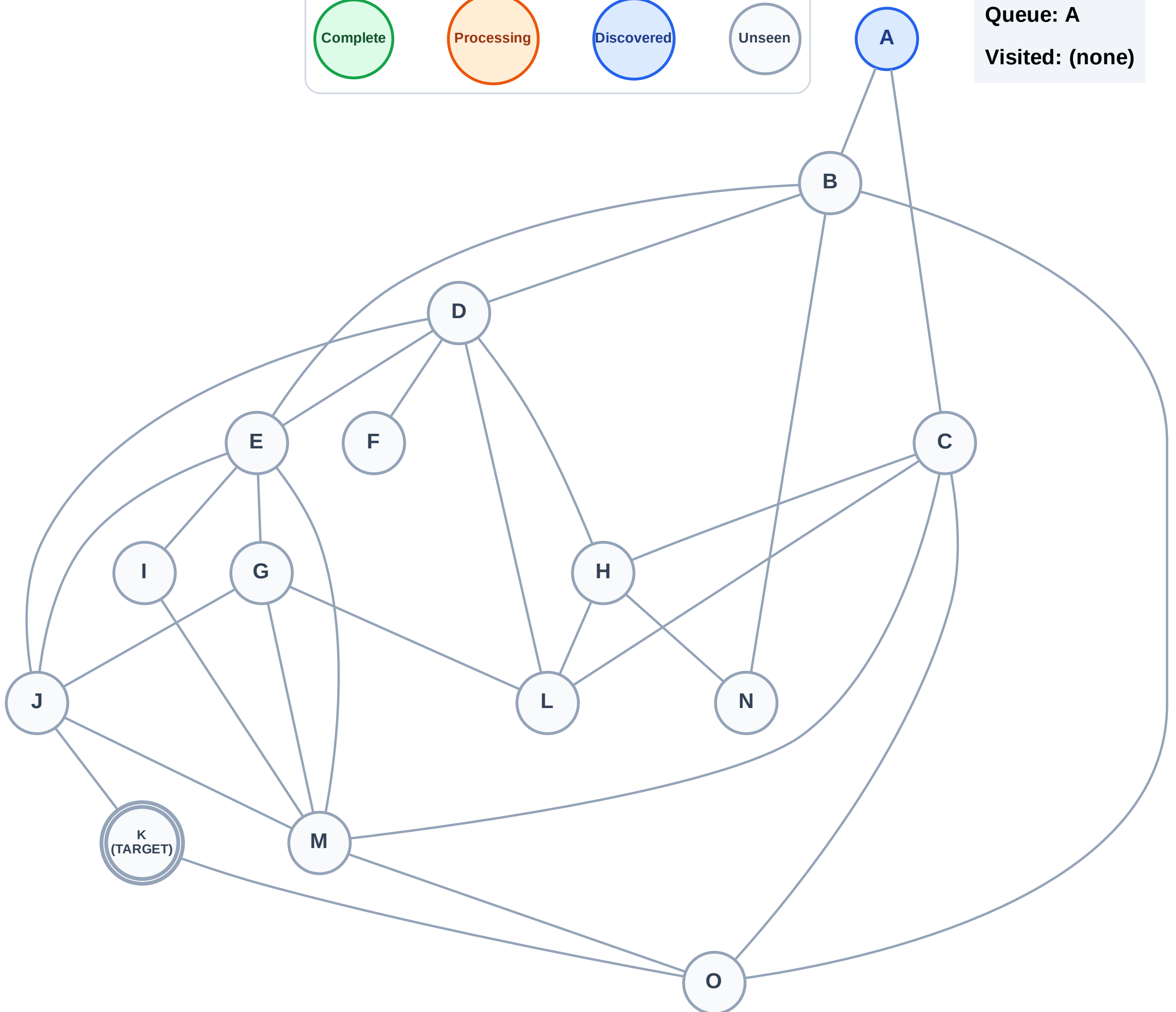
Discovered



Unseen

Queue: A

Visited: (none)



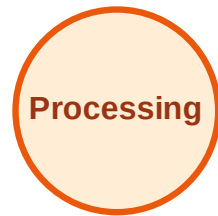
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



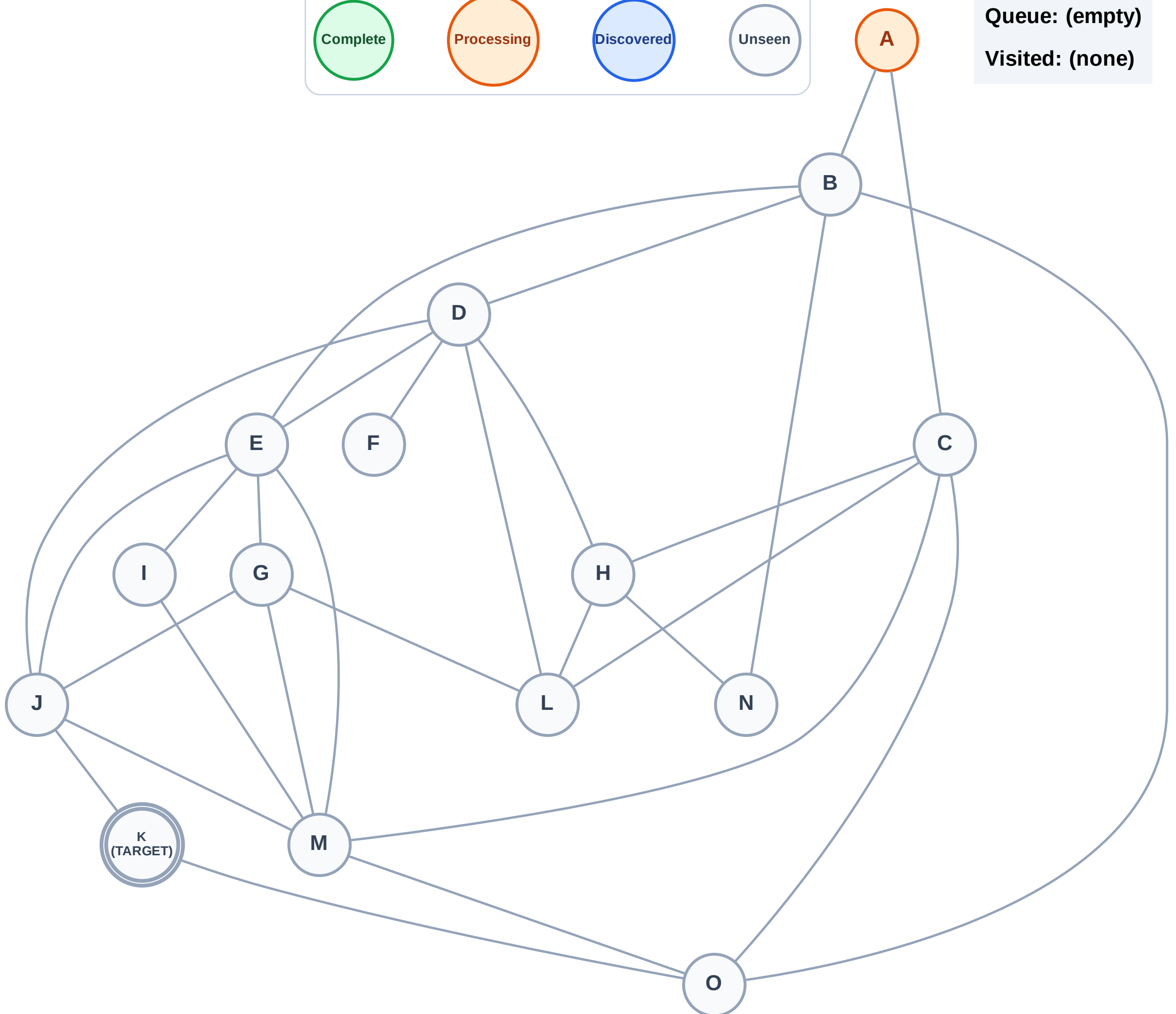
Discovered



Unseen

Queue: (empty)

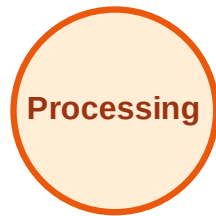
Visited: (none)



BFS Traversal

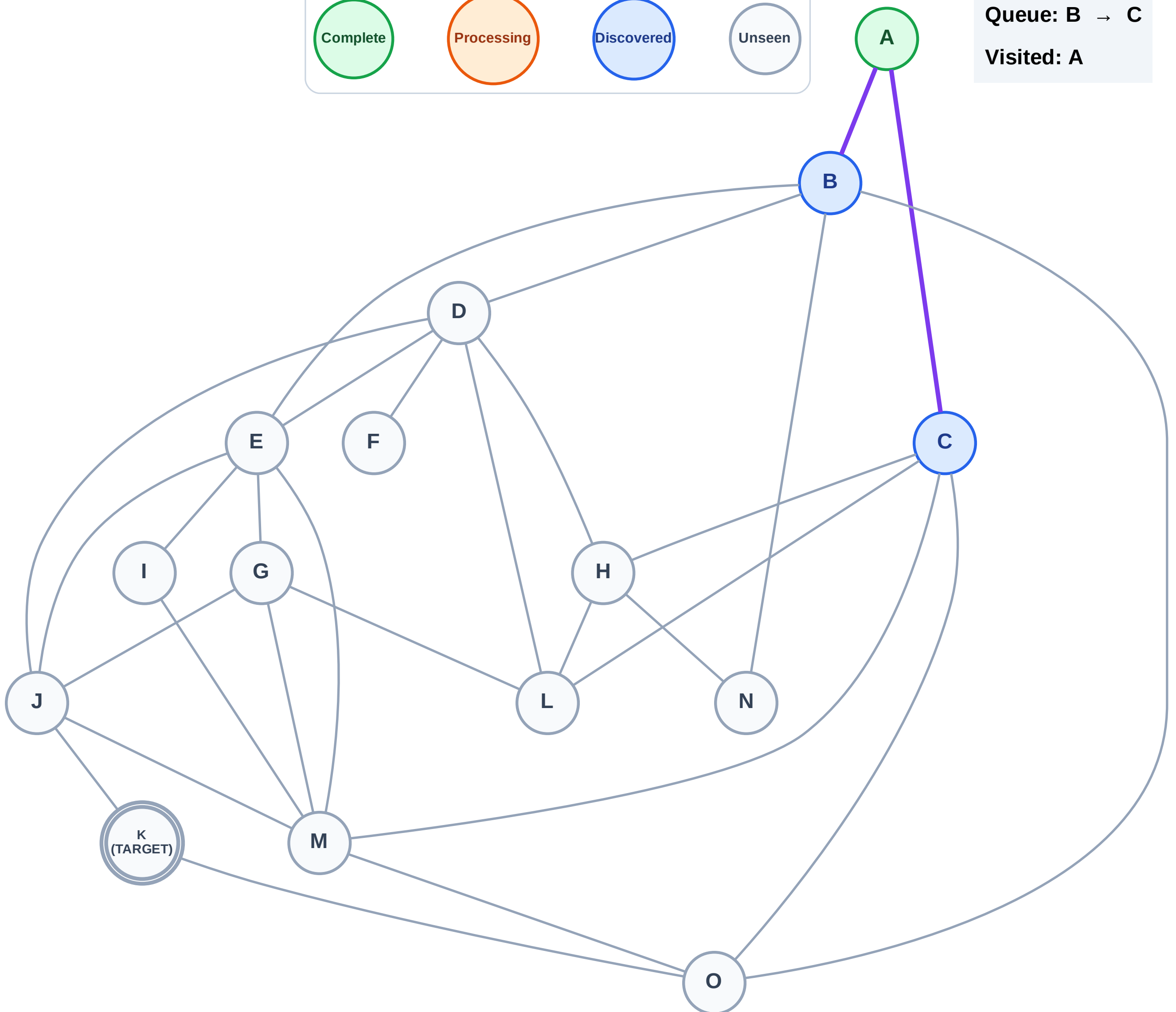
Step 3: Discover B, C from A

Legend



Queue: B → C

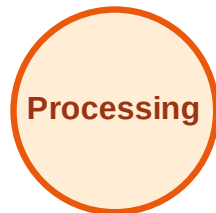
Visited: A



BFS Traversal

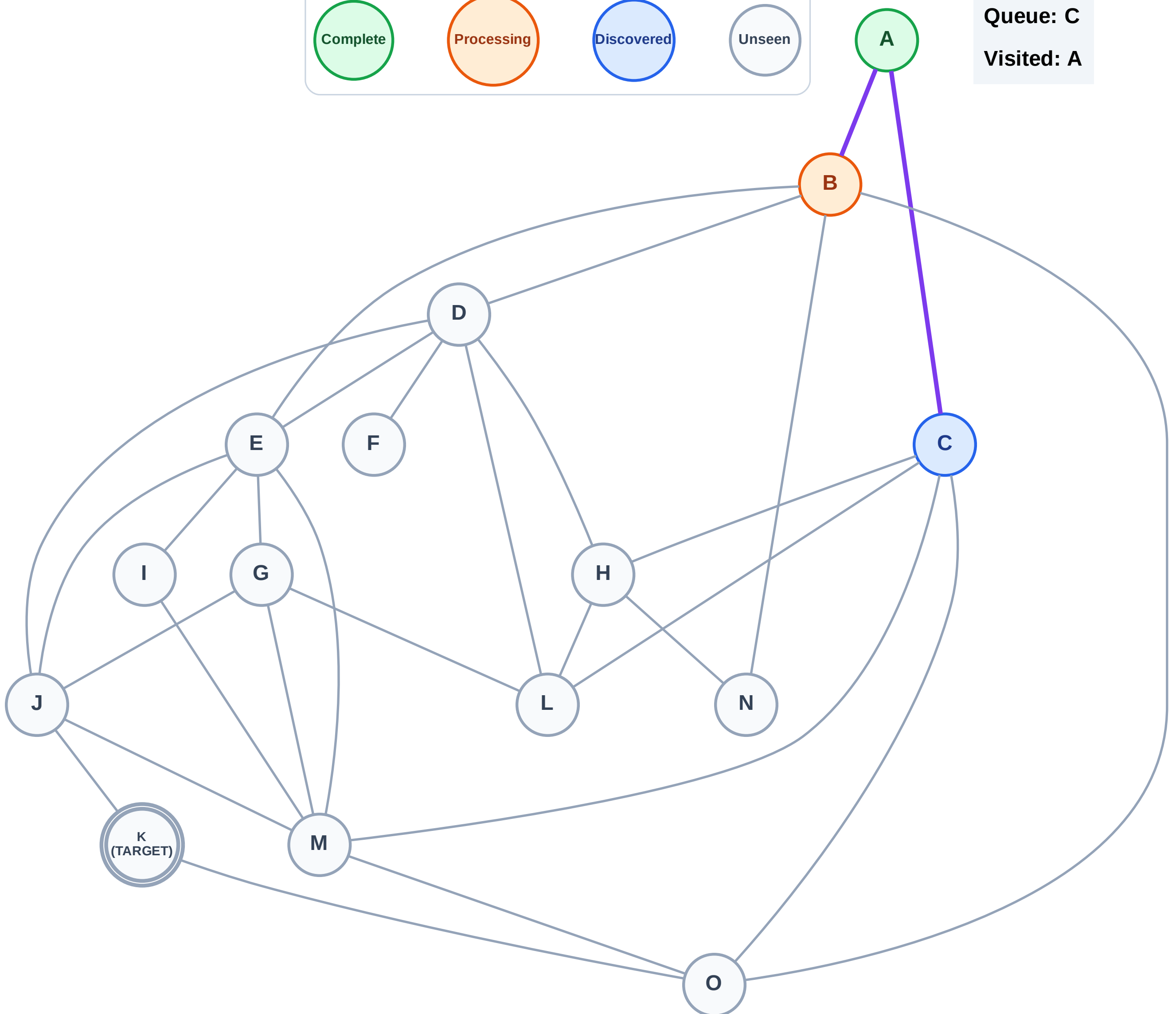
Step 4: Process node B

Legend



Queue: C

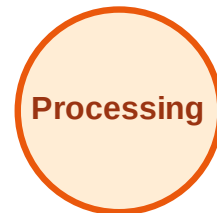
Visited: A



BFS Traversal

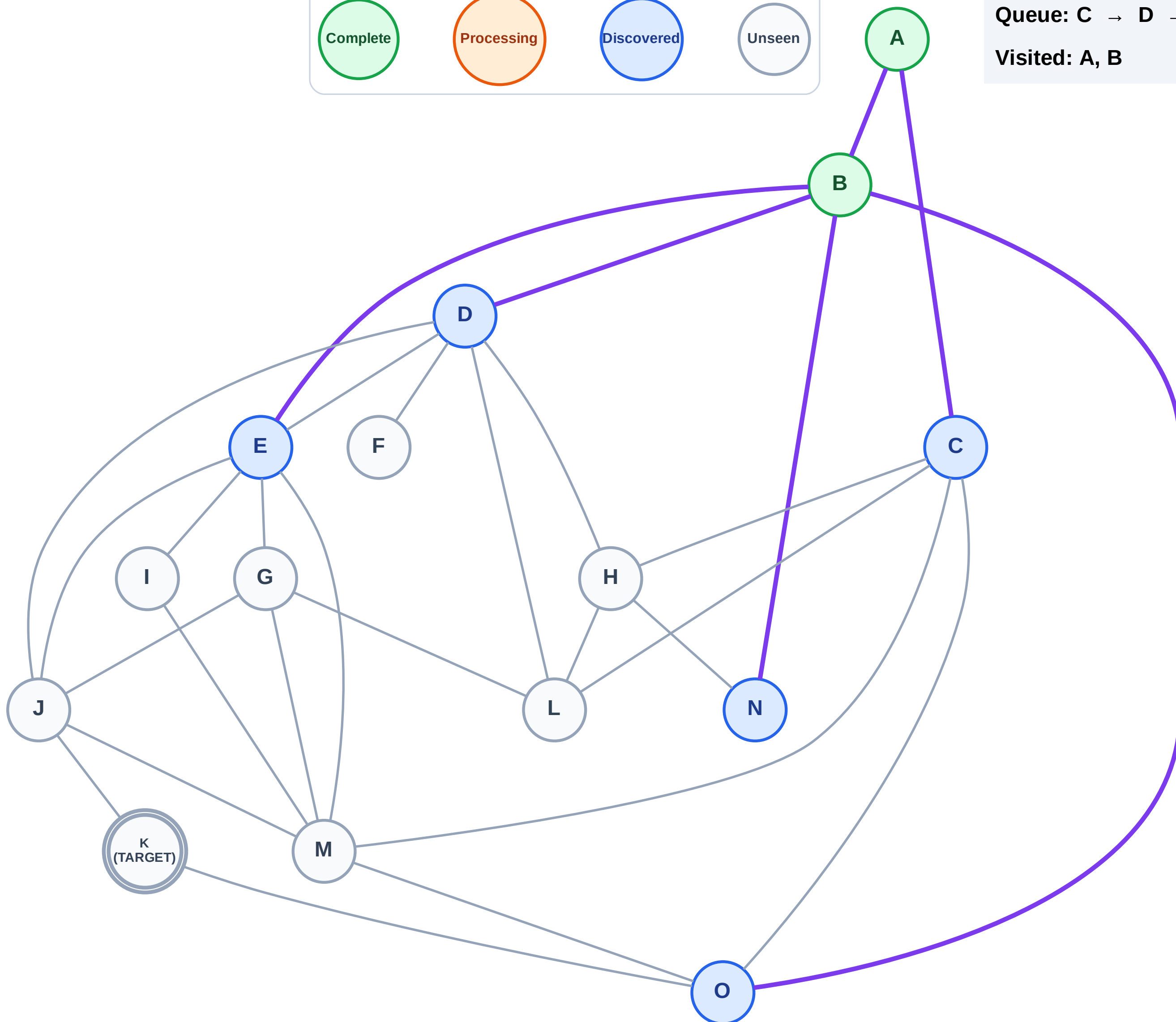
Step 5: Discover D, E, N, O from B

Legend



Queue: C → D → E → N → O

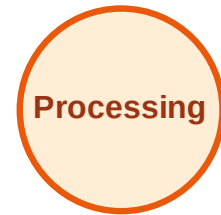
Visited: A, B



BFS Traversal

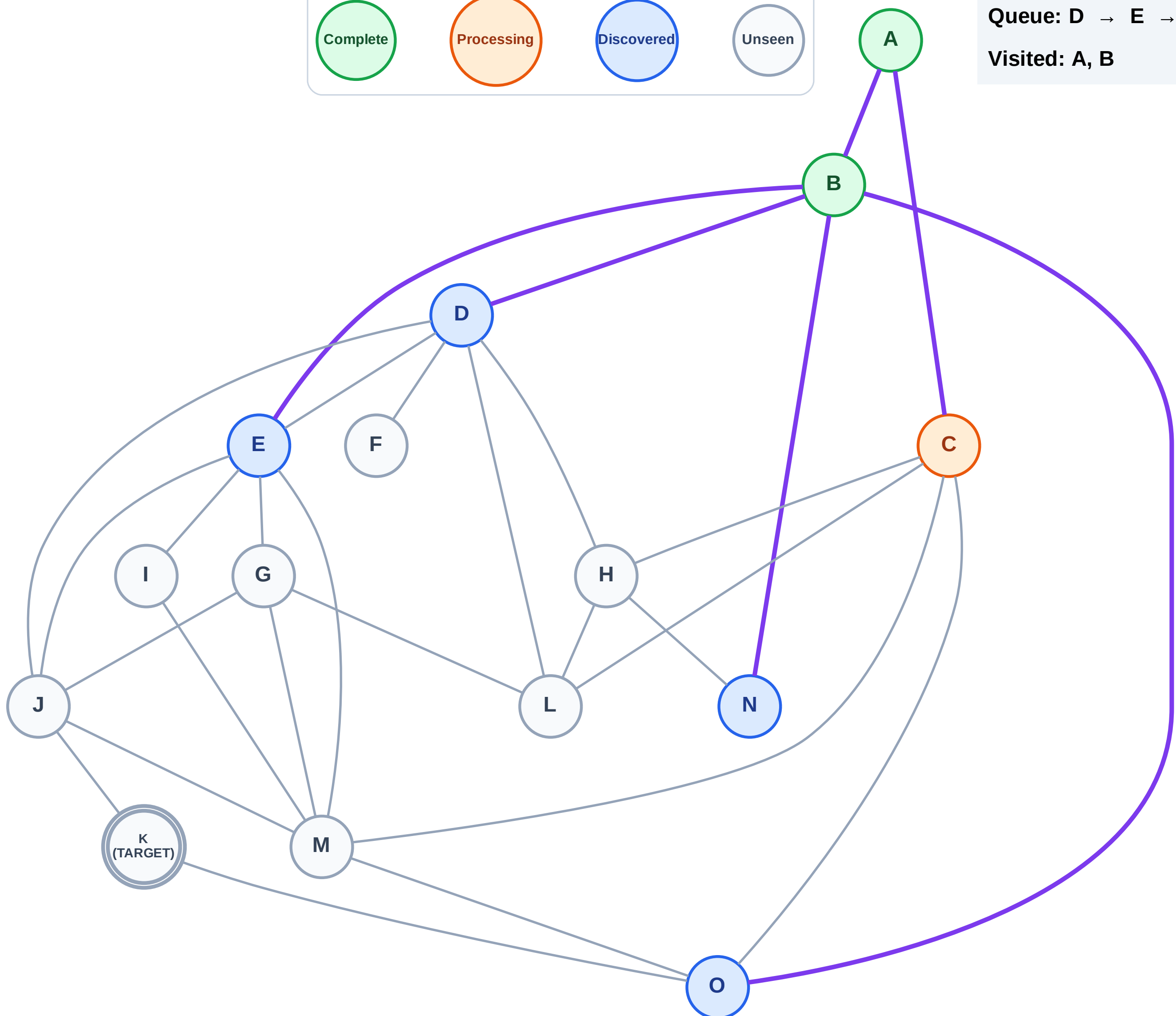
Step 6: Process node C

Legend



Queue: D → E → N → O

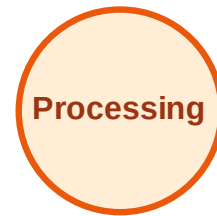
Visited: A, B



BFS Traversal

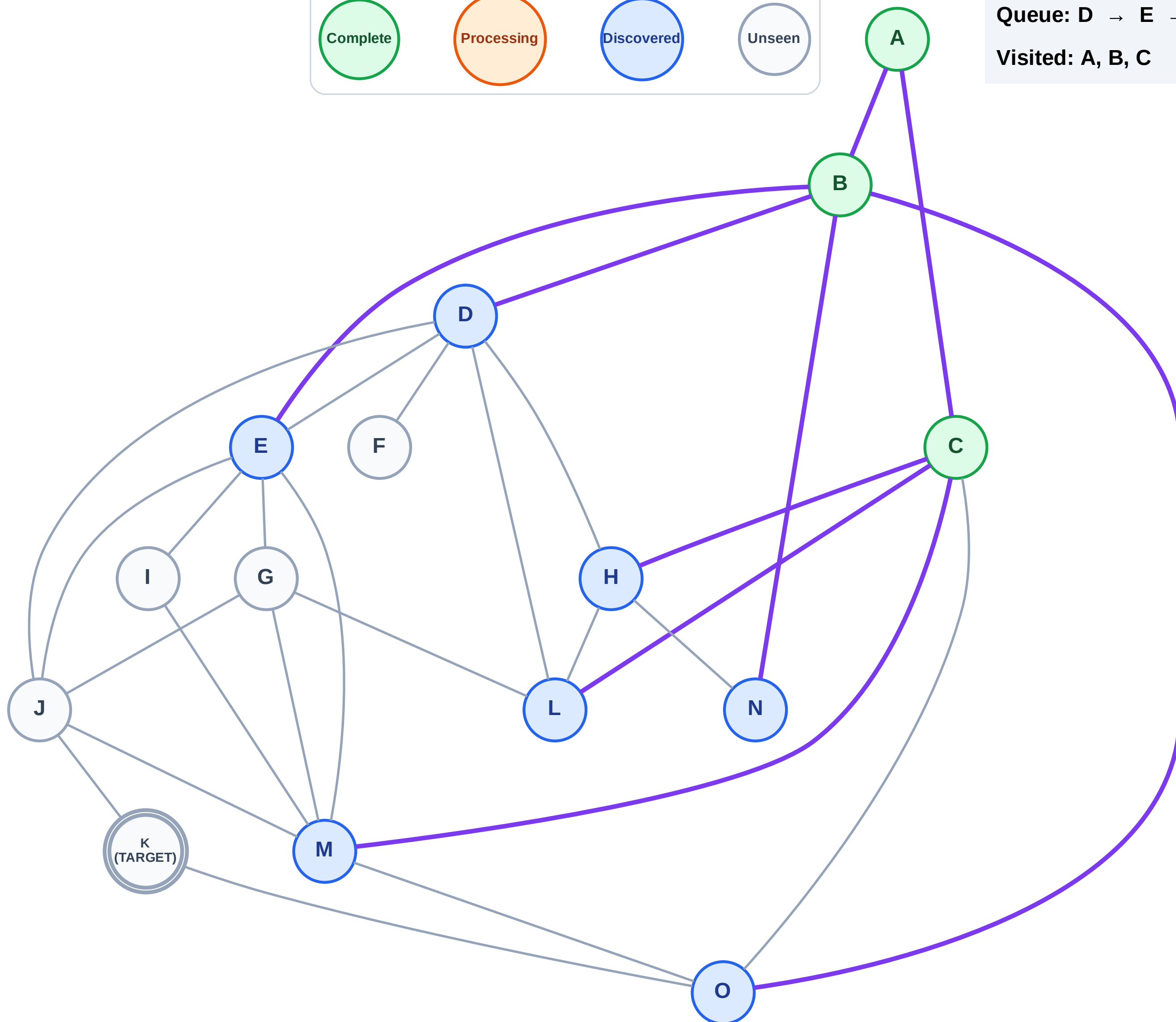
Step 7: Discover H, L, M from C

Legend



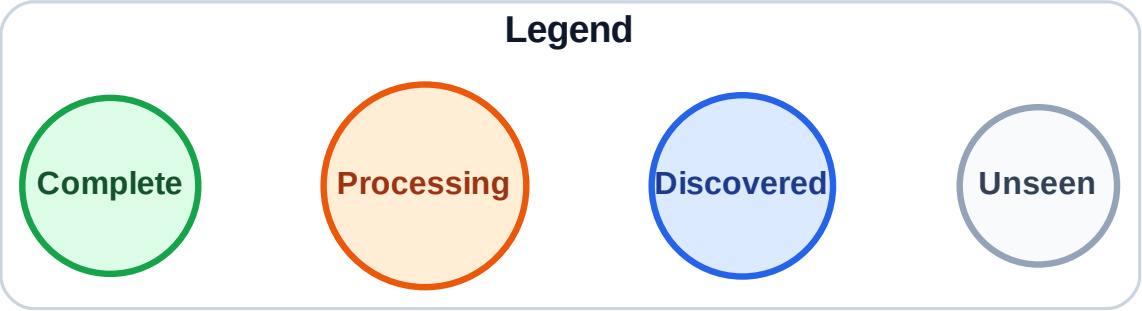
Queue: D → E → N → O → H → L → M

Visited: A, B, C

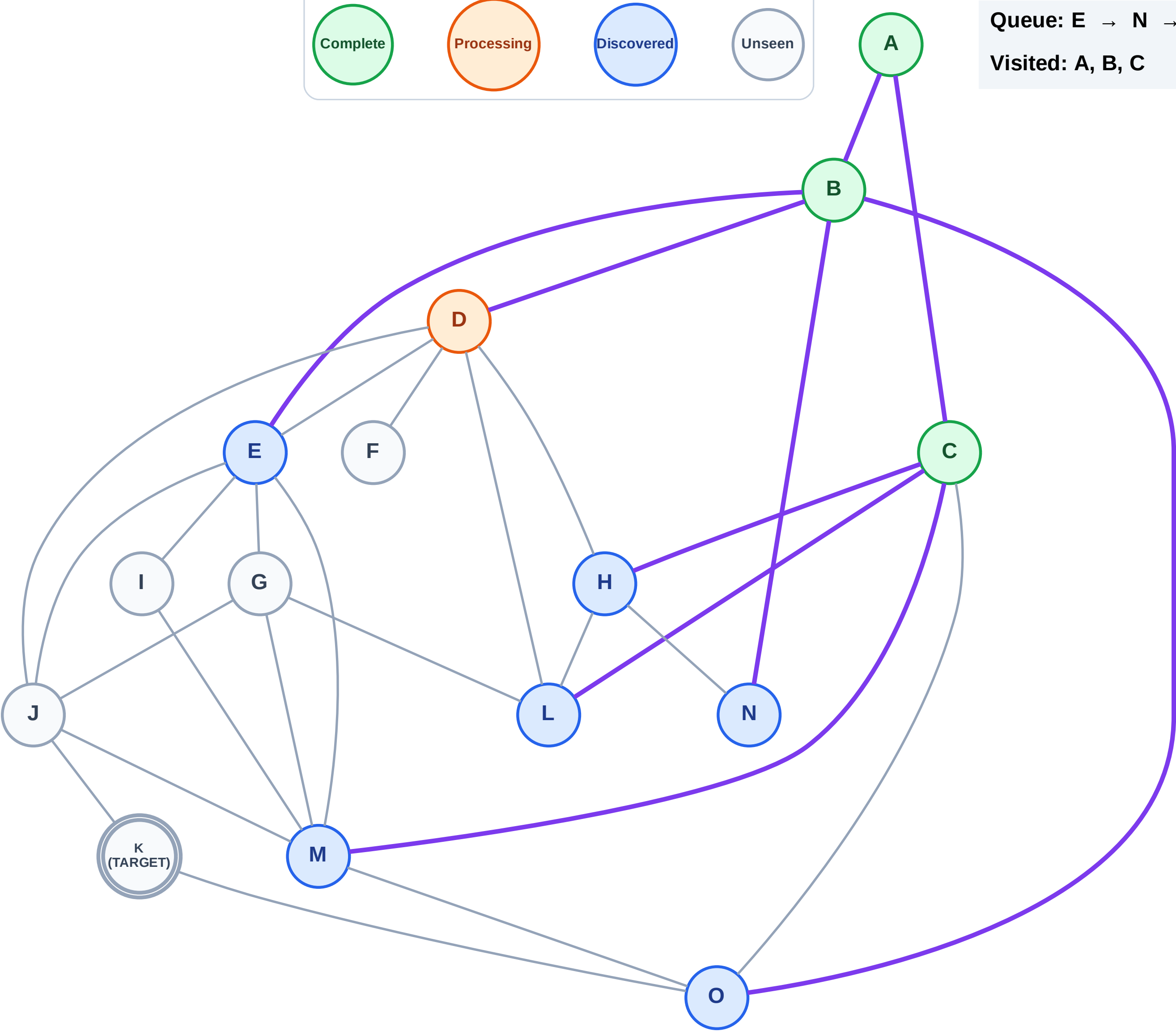


BFS Traversal

Step 8: Process node D



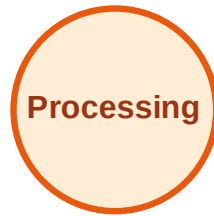
Queue: E → N → O → H → L → M
Visited: A, B, C



BFS Traversal

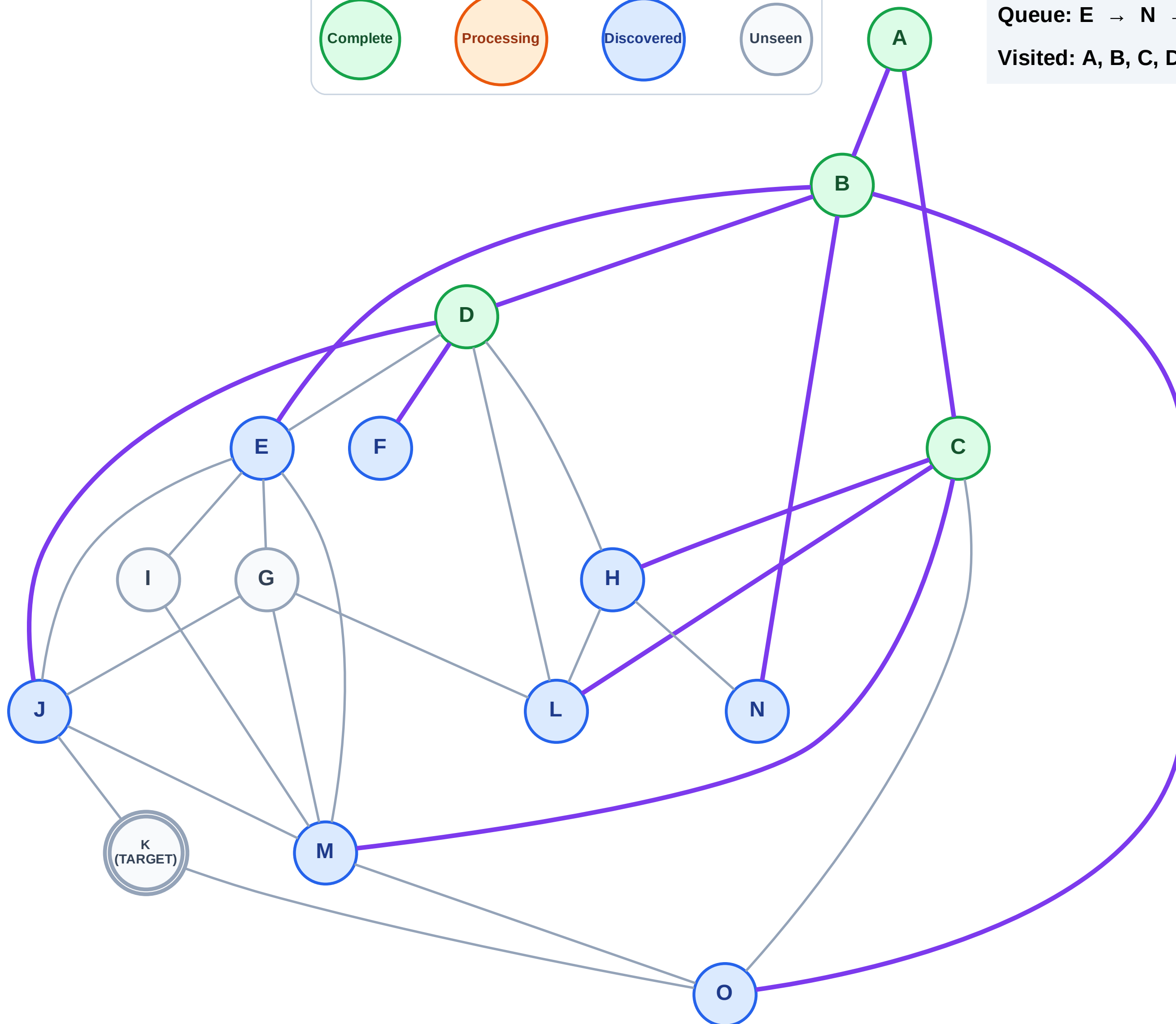
Step 9: Discover F, J from D

Legend



Queue: E → N → O → H → L → M → F → J

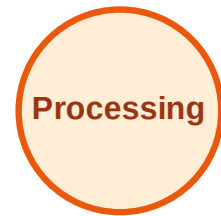
Visited: A, B, C, D



BFS Traversal

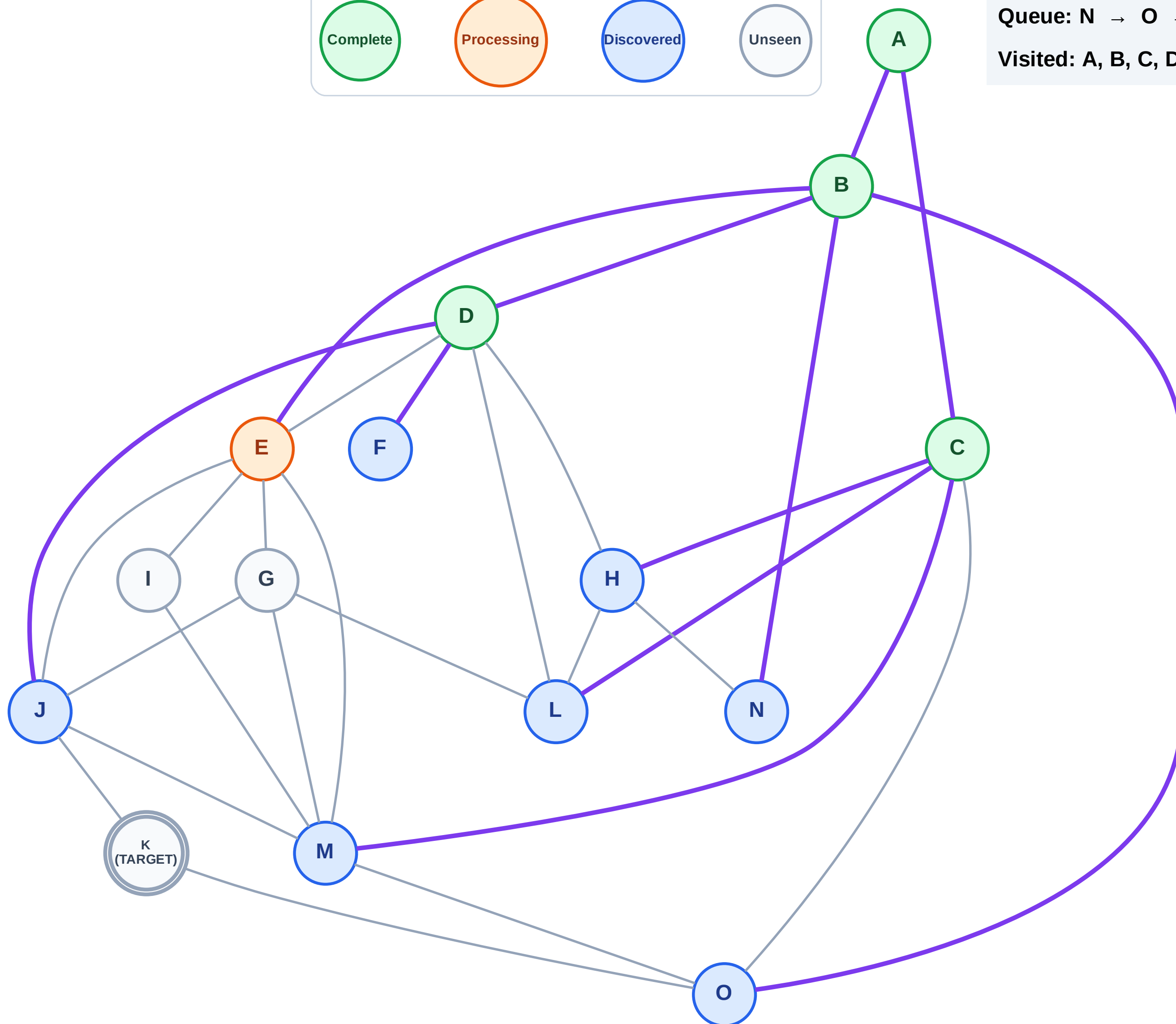
Step 10: Process node E

Legend



Queue: N → O → H → L → M → F → J

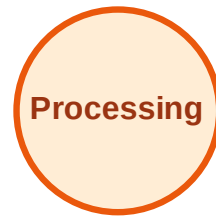
Visited: A, B, C, D



BFS Traversal

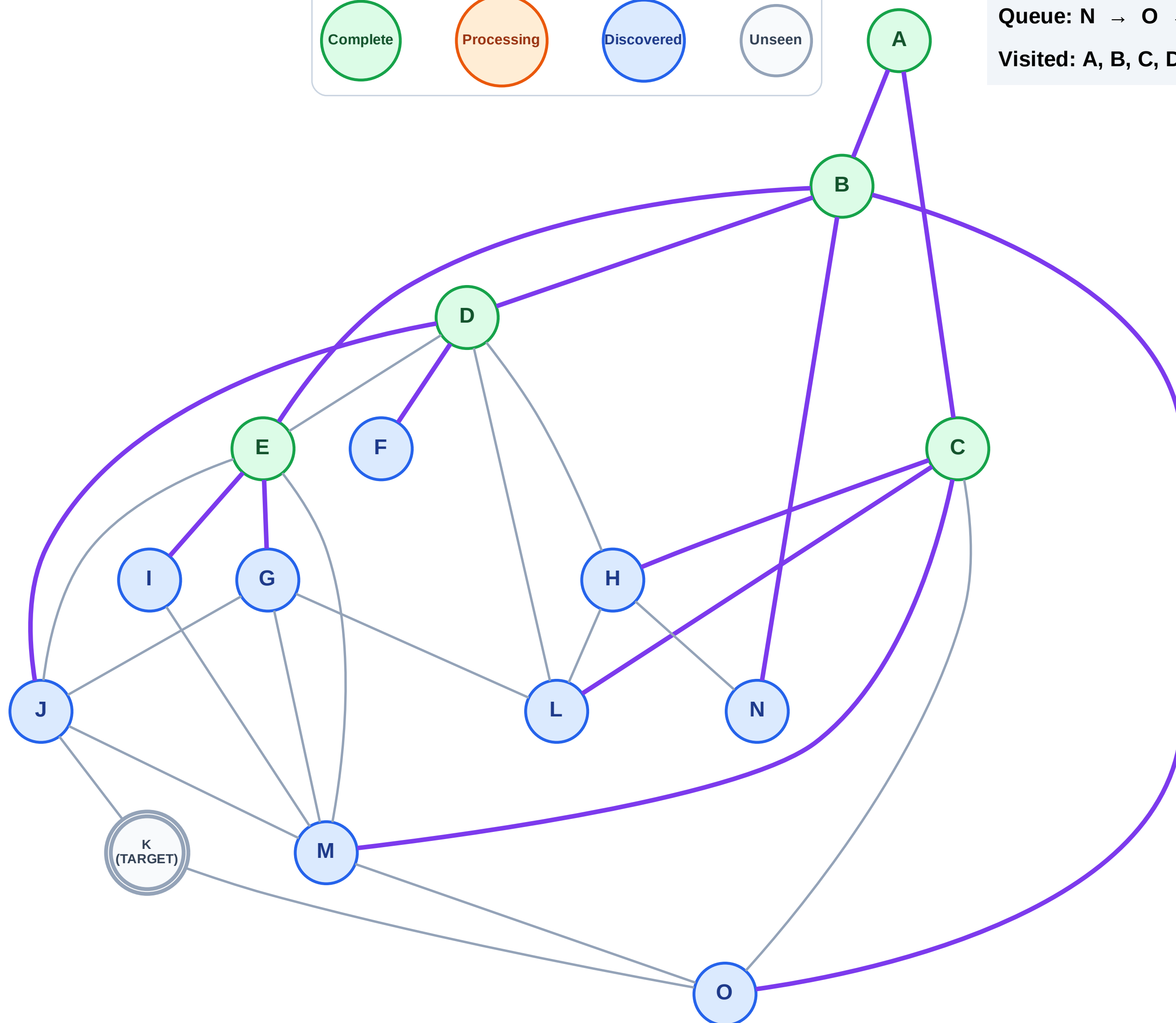
Step 11: Discover G, I from E

Legend



Queue: N → O → H → L → M → F → J → G → I

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node N

Legend

Complete

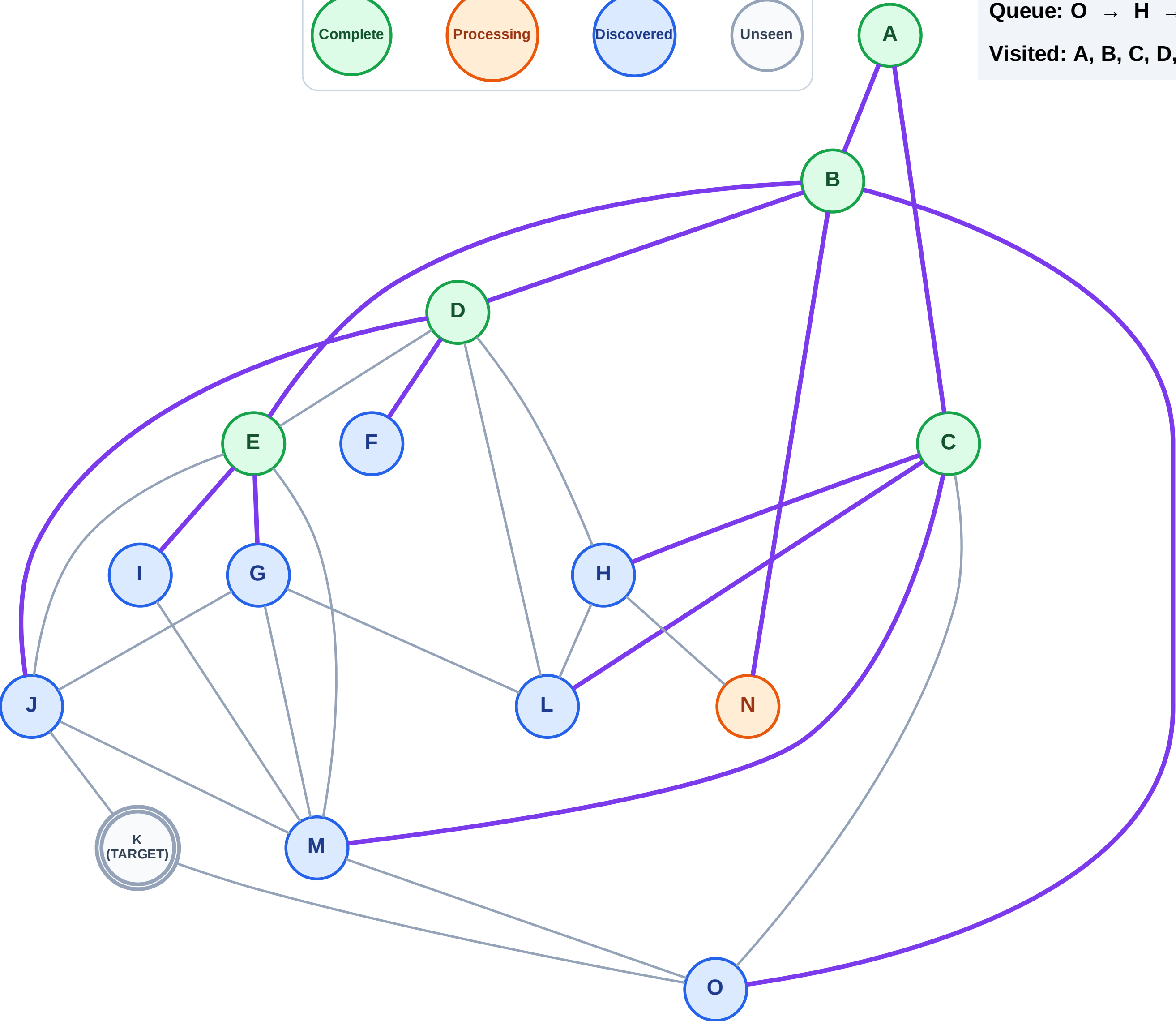
Processing

Discovered

Unseen

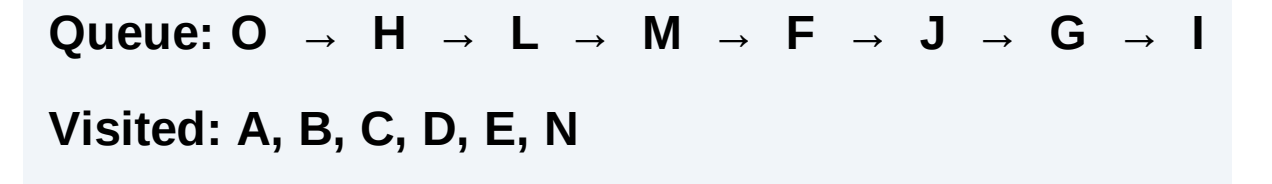
Queue: O → H → L → M → F → J → G → I

Visited: A, B, C, D, E



Step 13: No new neighbors from N

Step 13: No new neighbors from N



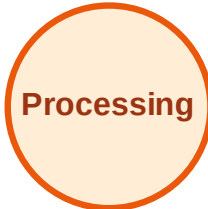
BFS Traversal

Step 14: Process node O

Legend



Complete



Processing



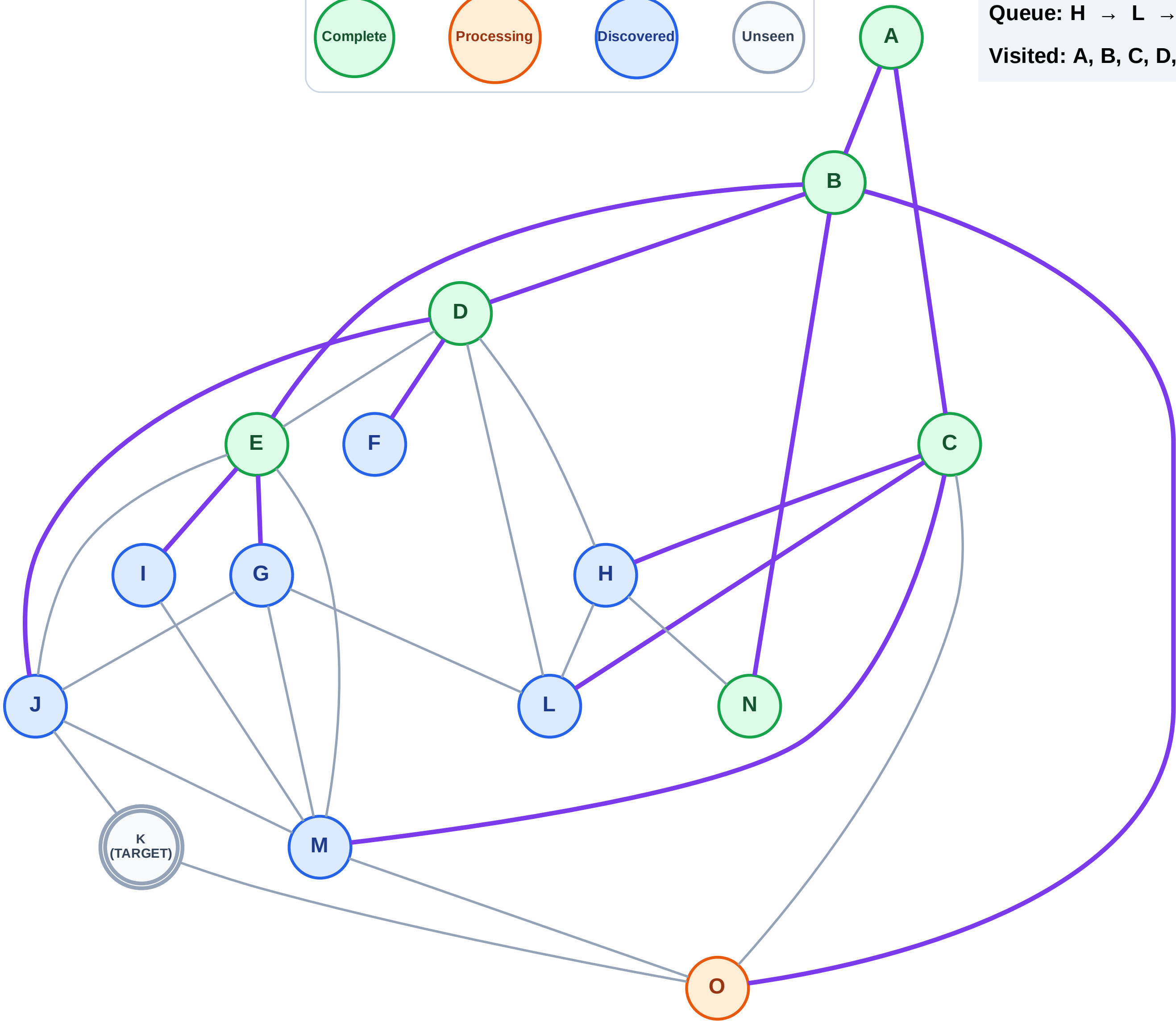
Discovered



Unseen

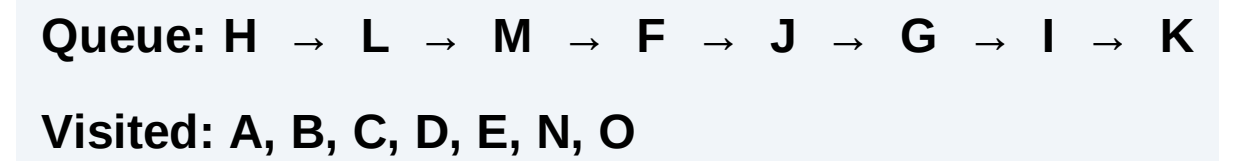
Queue: H → L → M → F → J → G → I

Visited: A, B, C, D, E, N



Step 15: Discover K from O

Step 15: Discover K from O



BFS Traversal

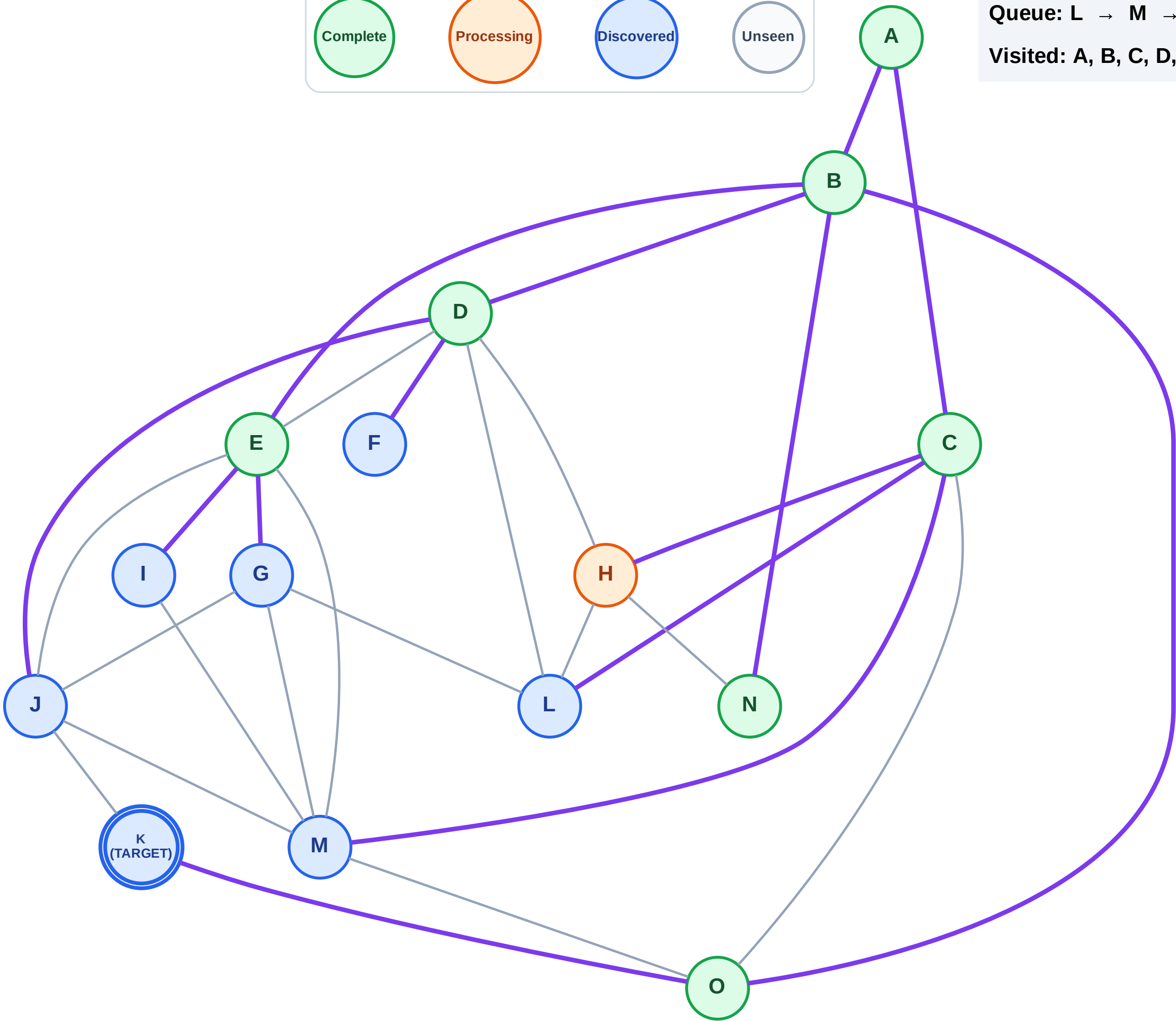
Step 16: Process node H

Legend



Queue: L → M → F → J → G → I → K

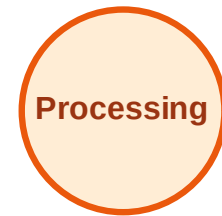
Visited: A, B, C, D, E, N, O



BFS Traversal

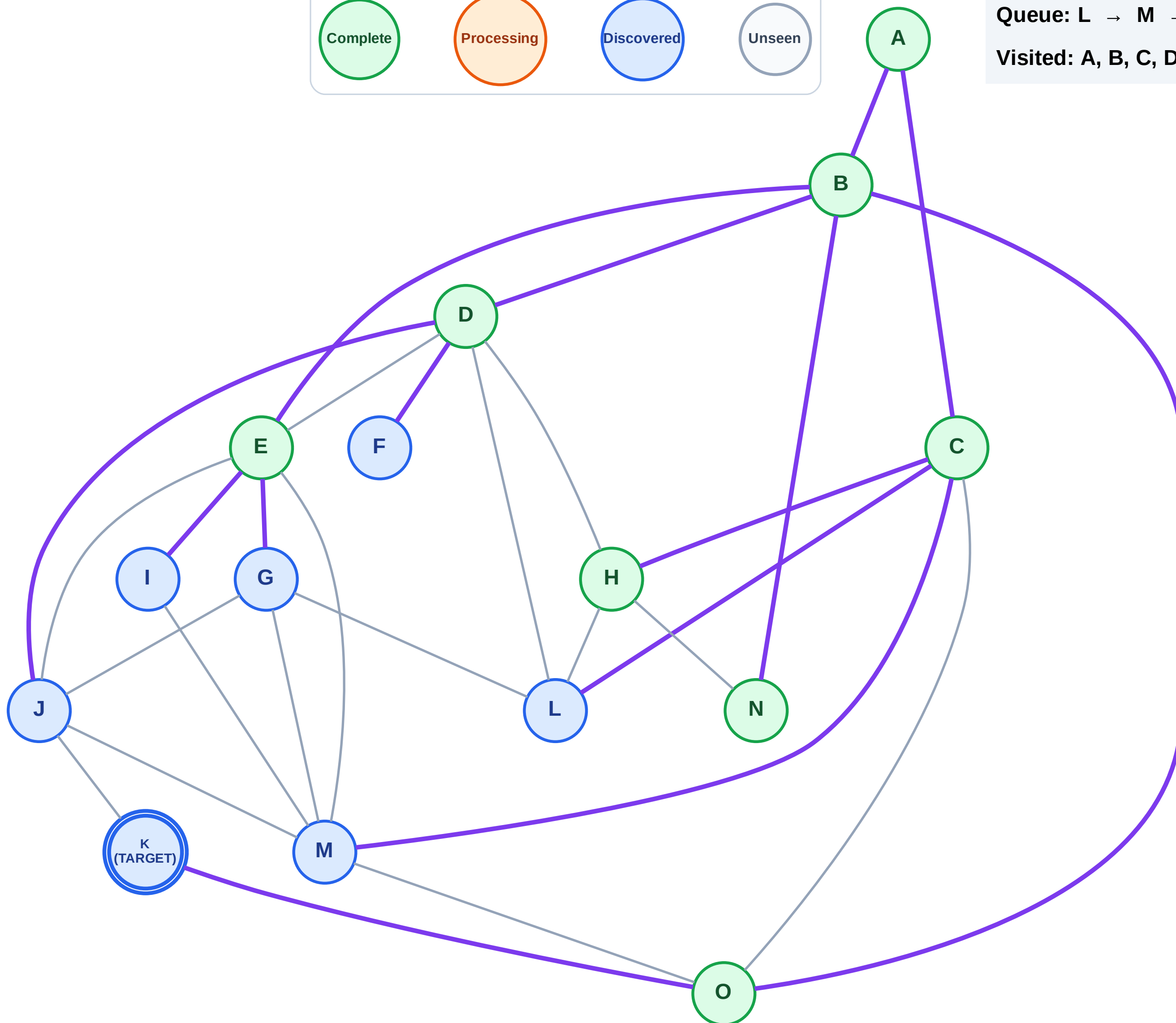
Step 17: No new neighbors from H

Legend



Queue: L → M → F → J → G → I → K

Visited: A, B, C, D, E, H, N, O



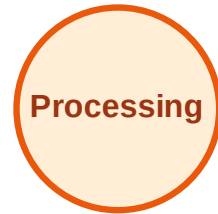
BFS Traversal

Step 18: Process node L

Legend



Complete



Processing



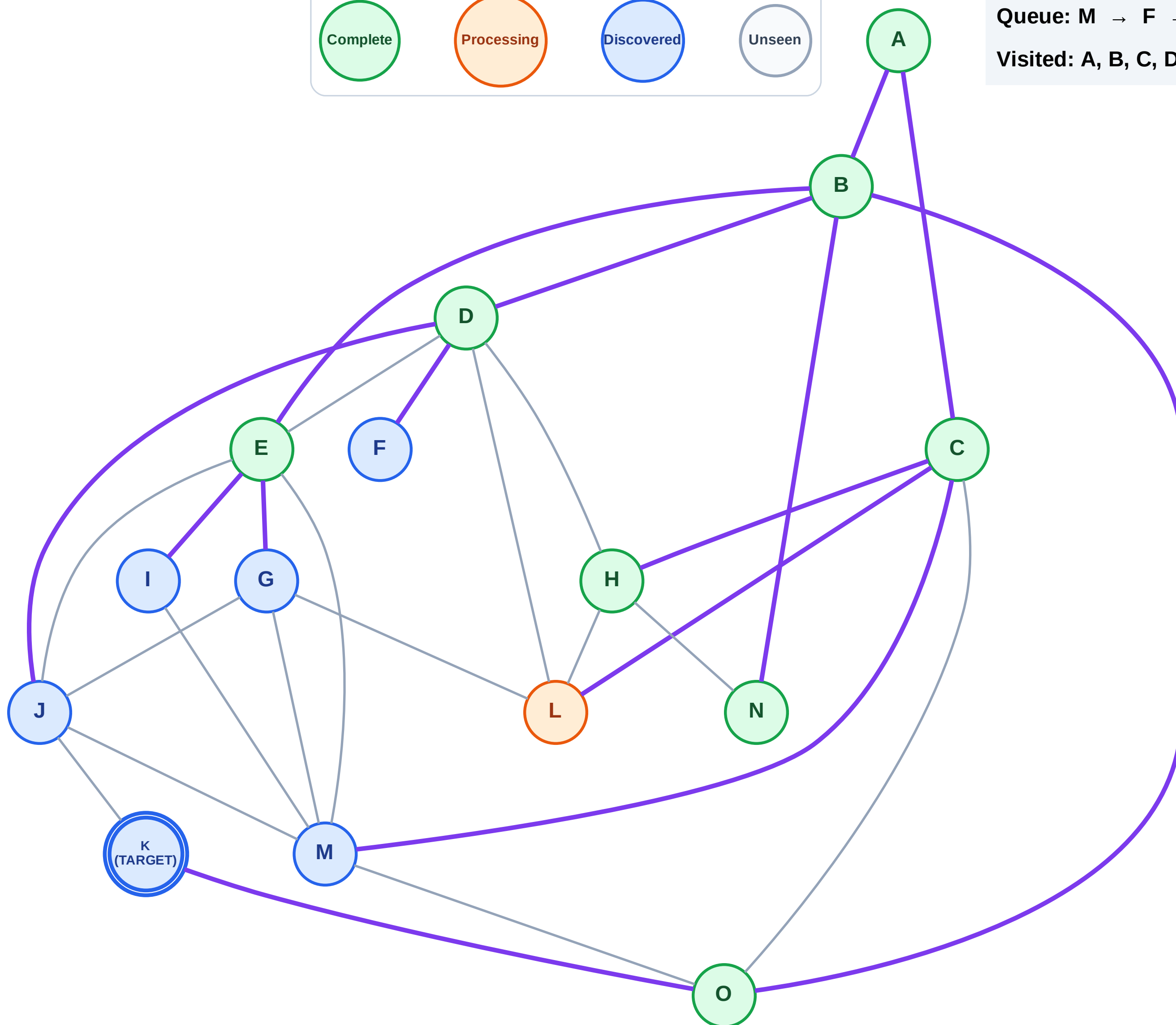
Discovered



Unseen

Queue: M → F → J → G → I → K

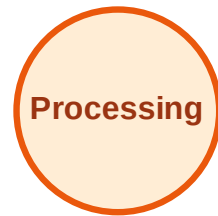
Visited: A, B, C, D, E, H, N, O



BFS Traversal

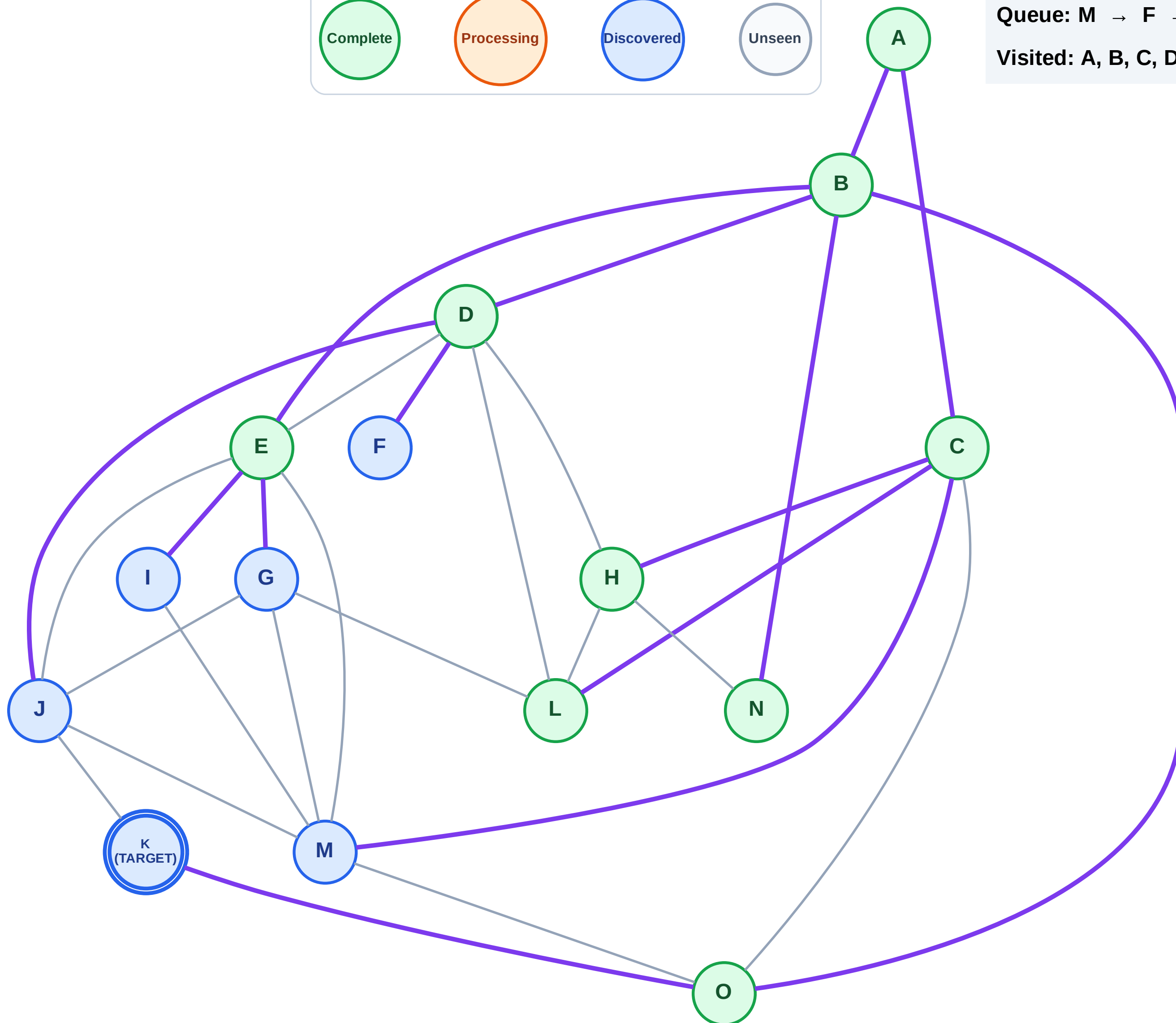
Step 19: No new neighbors from L

Legend



Queue: M → F → J → G → I → K

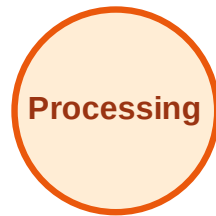
Visited: A, B, C, D, E, H, L, N, O



BFS Traversal

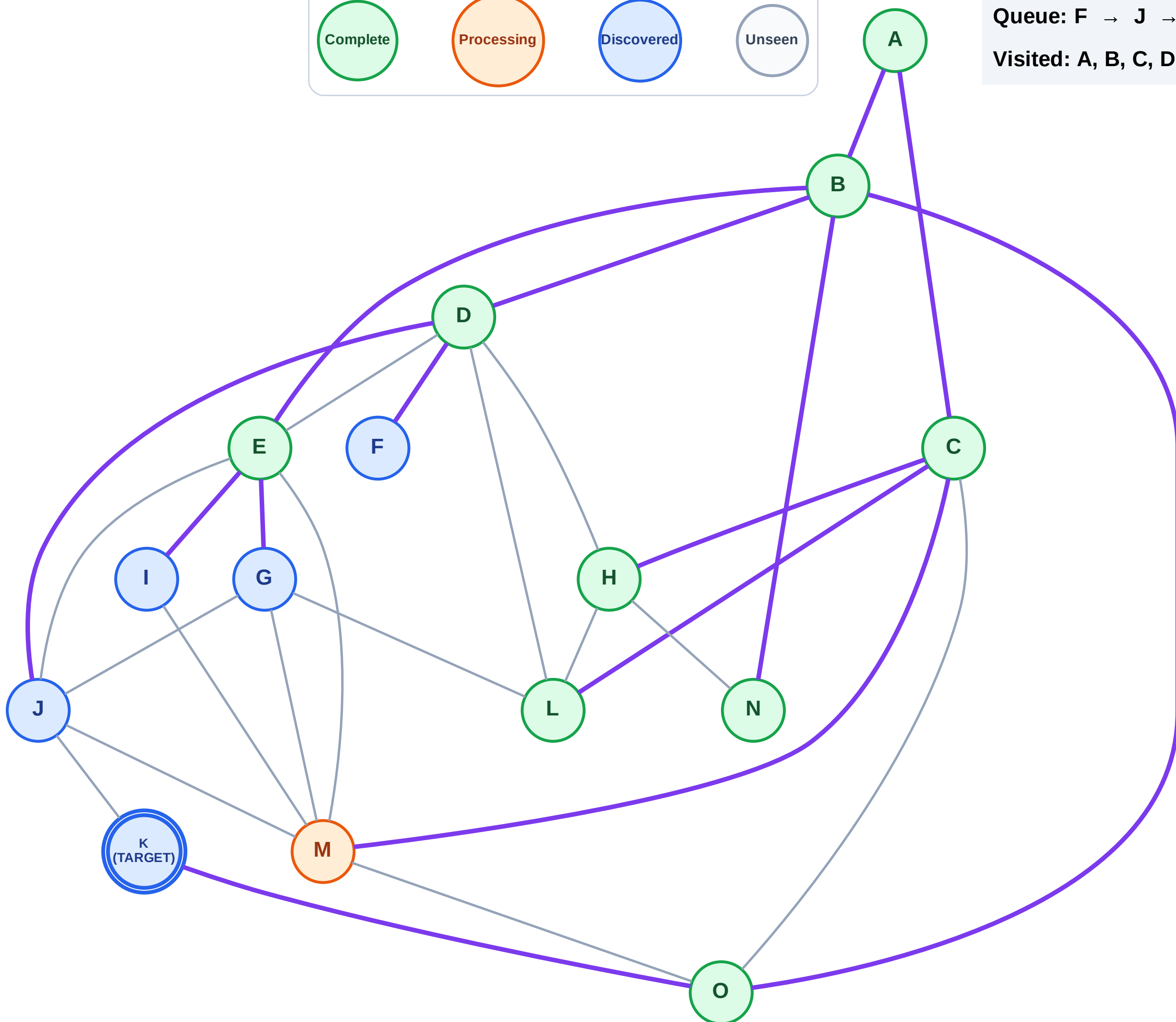
Step 20: Process node M

Legend



Queue: F → J → G → I → K

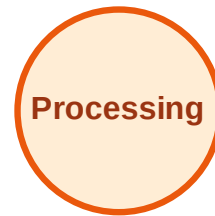
Visited: A, B, C, D, E, H, L, N, O



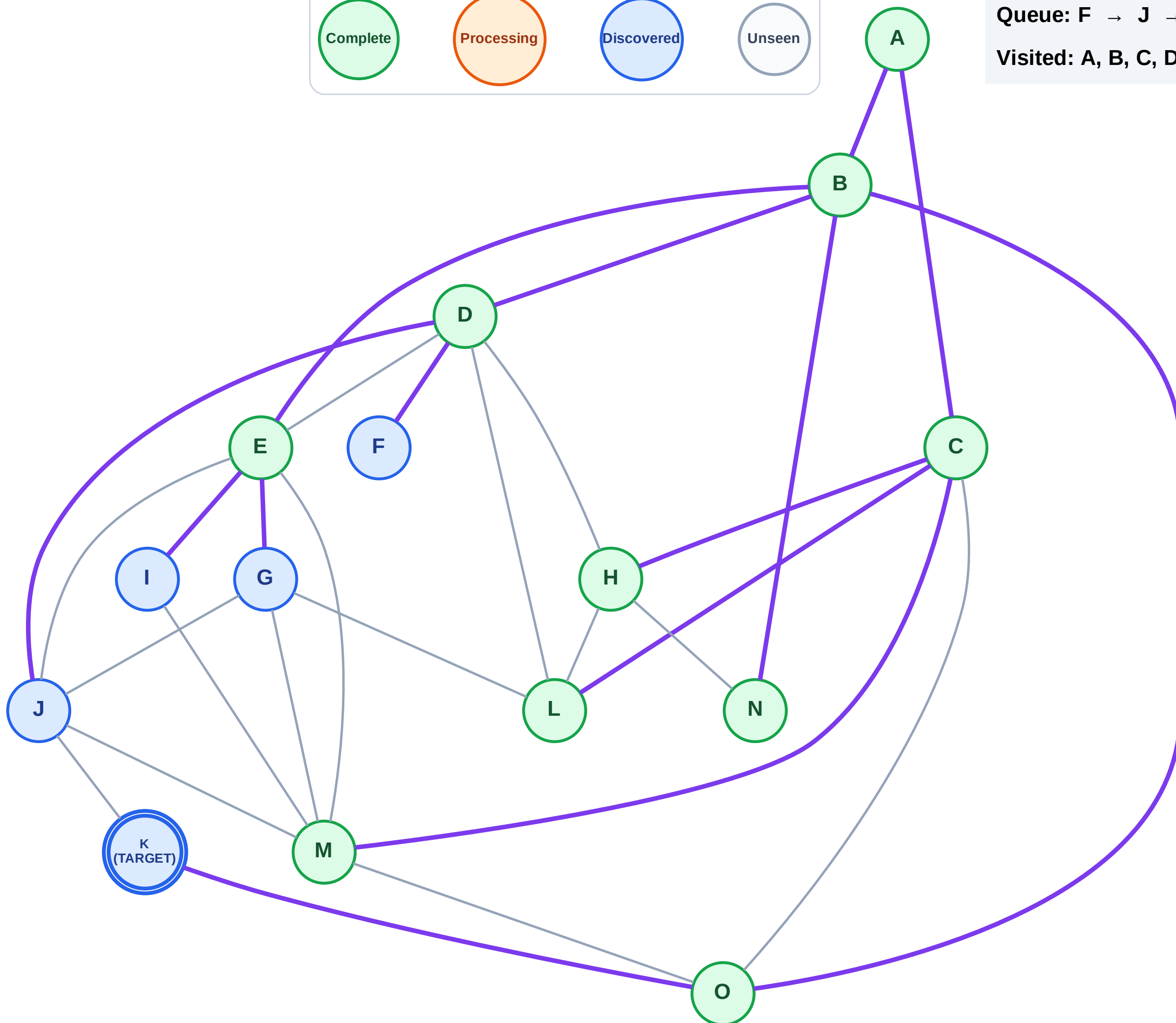
BFS Traversal

Step 21: No new neighbors from M

Legend



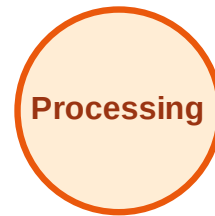
Queue: F → J → G → I → K
Visited: A, B, C, D, E, H, L, M, N, O



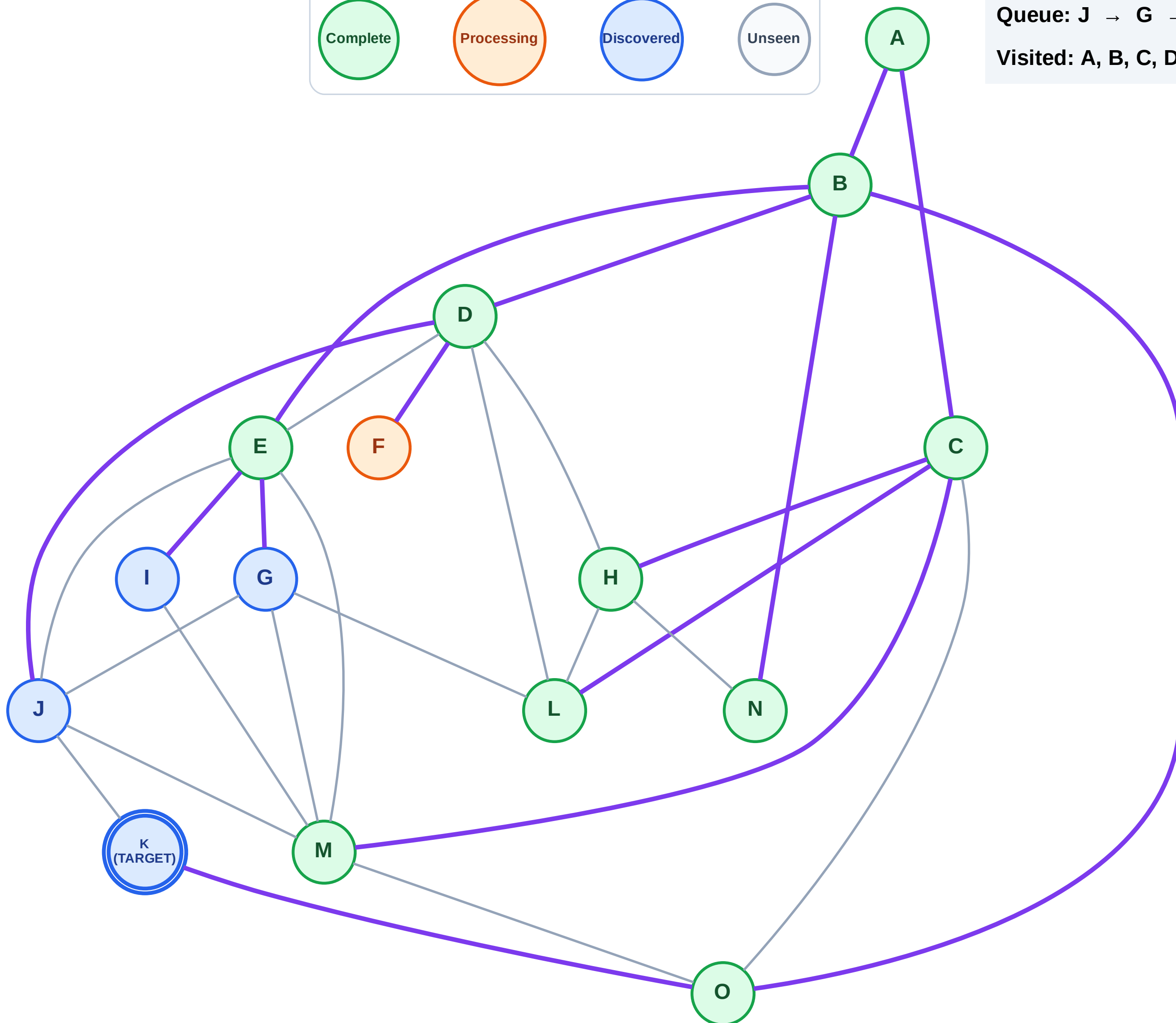
BFS Traversal

Step 22: Process node F

Legend



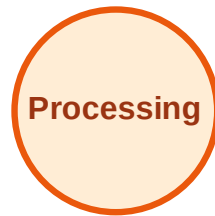
Queue: J → G → I → K
Visited: A, B, C, D, E, H, L, M, N, O



BFS Traversal

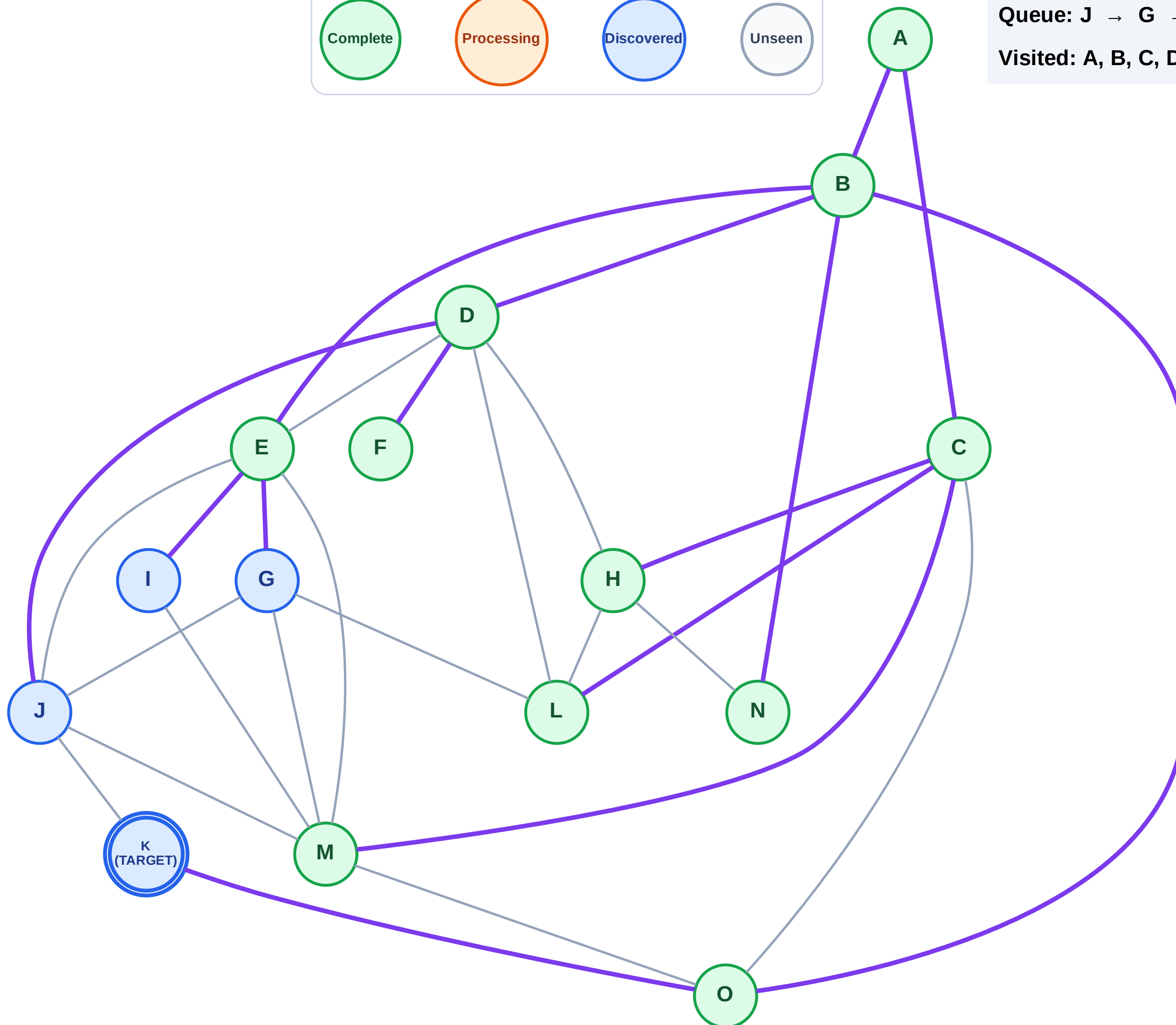
Step 23: No new neighbors from F

Legend



Queue: J → G → I → K

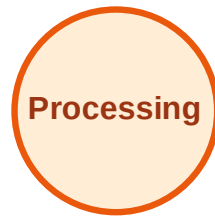
Visited: A, B, C, D, E, F, H, L, M, N, O



BFS Traversal

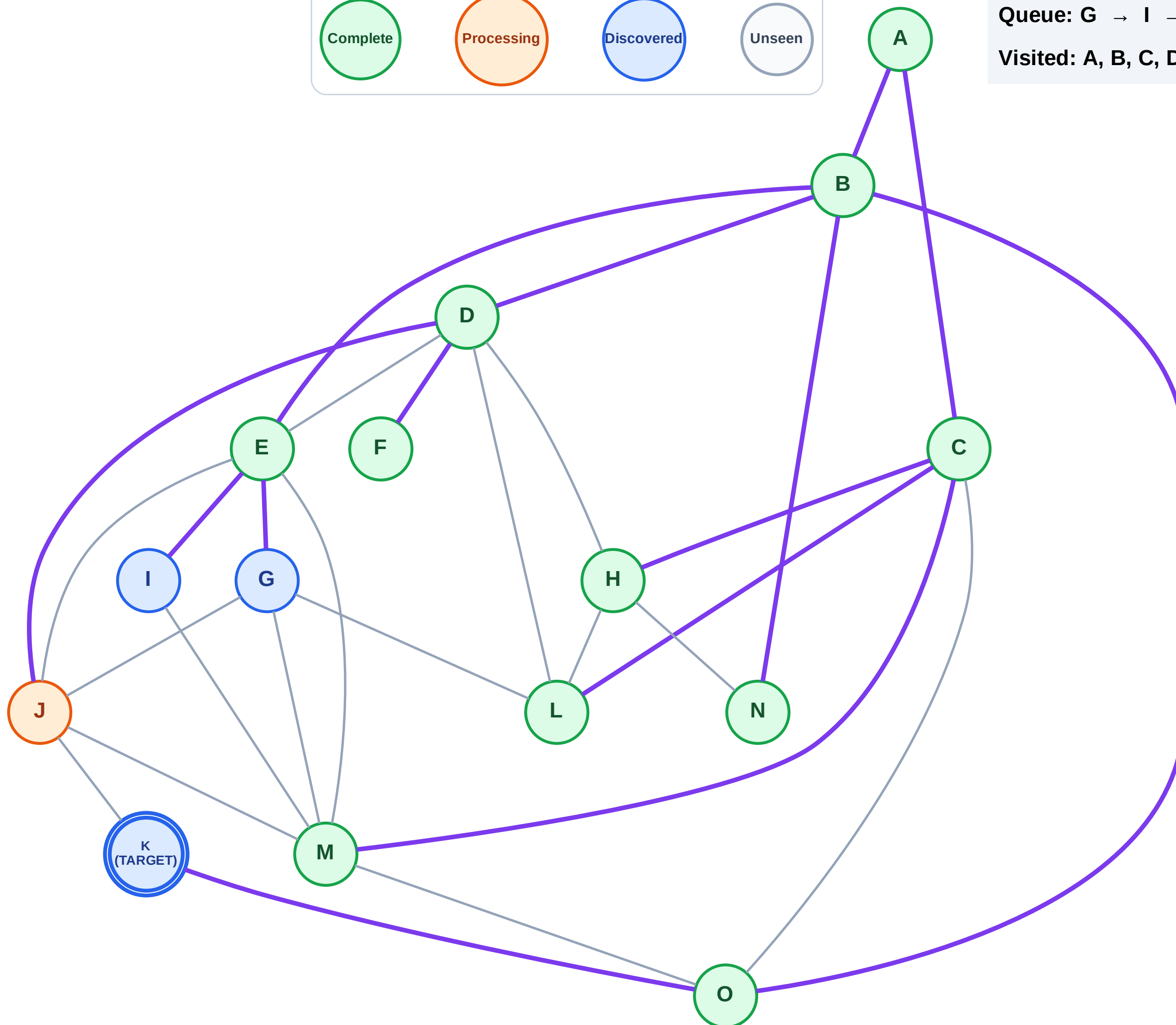
Step 24: Process node J

Legend



Queue: G → I → K

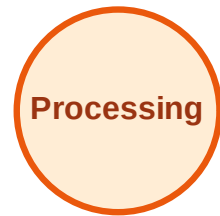
Visited: A, B, C, D, E, F, H, L, M, N, O



BFS Traversal

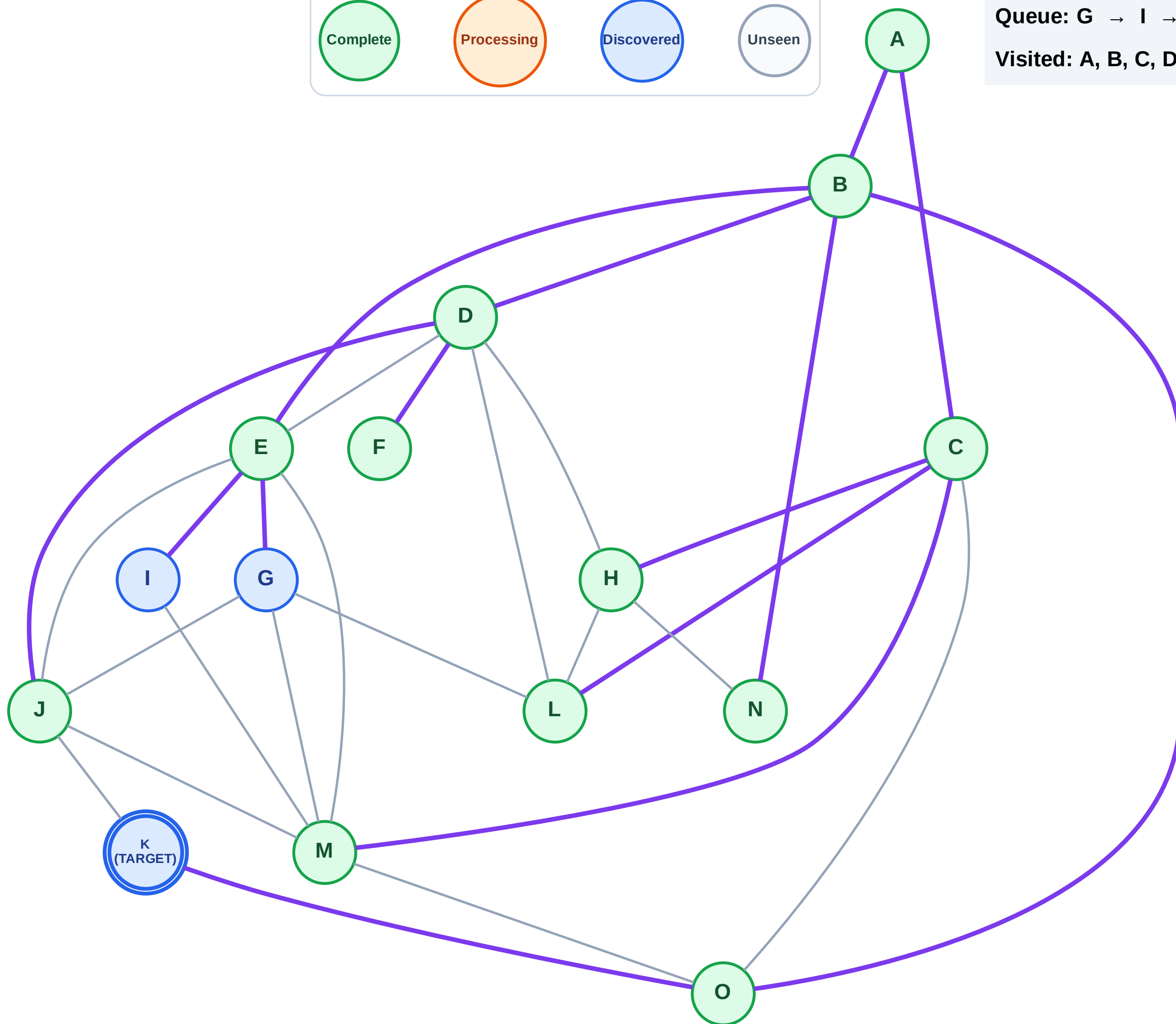
Step 25: No new neighbors from J

Legend



Queue: G → I → K

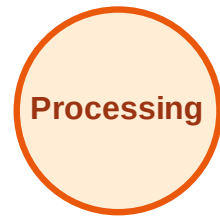
Visited: A, B, C, D, E, F, H, J, L, M, N, O



BFS Traversal

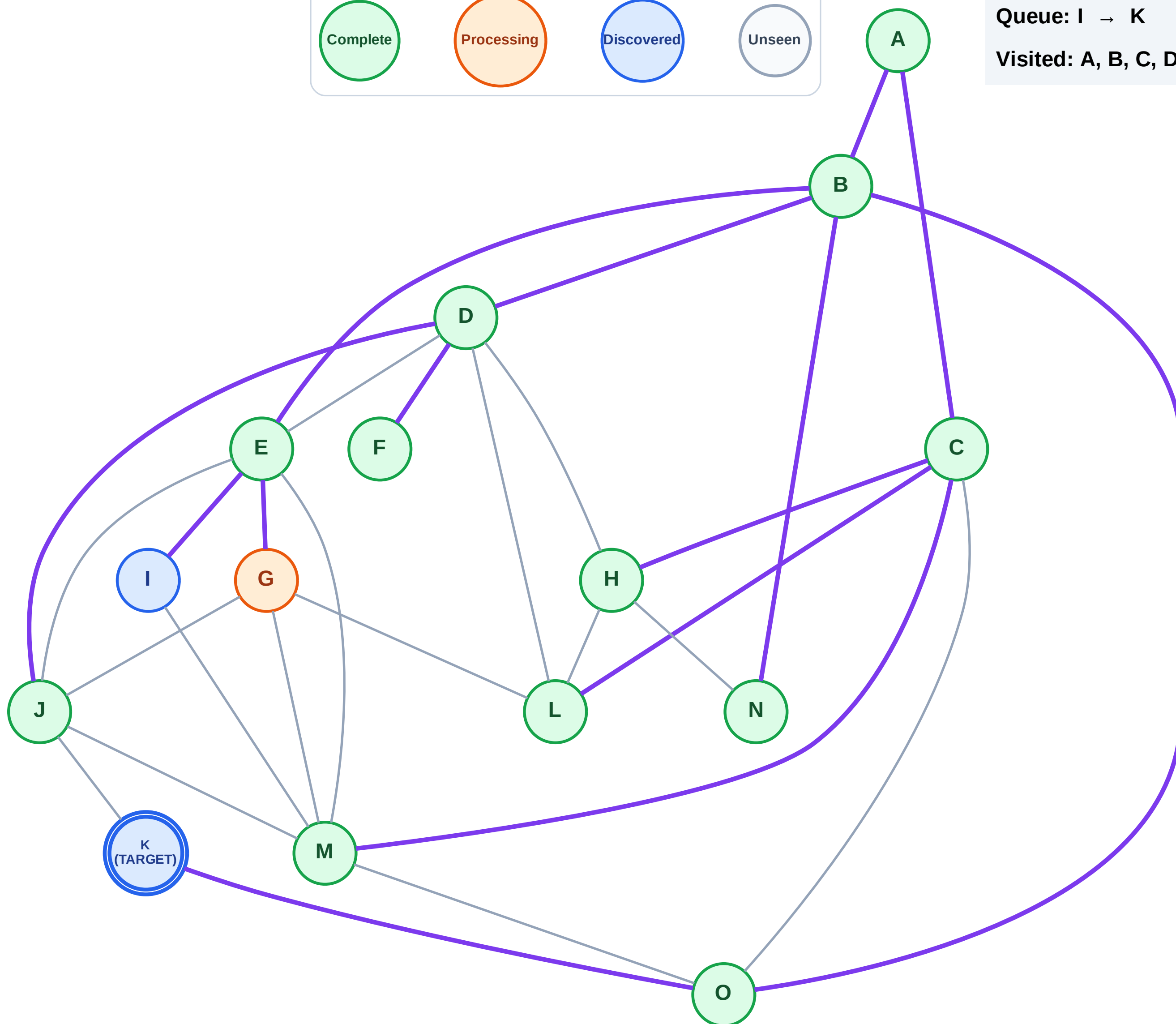
Step 26: Process node G

Legend



Queue: I → K

Visited: A, B, C, D, E, F, H, J, L, M, N, O



BFS Traversal

Step 27: No new neighbors from G

Legend



Complete



Processing



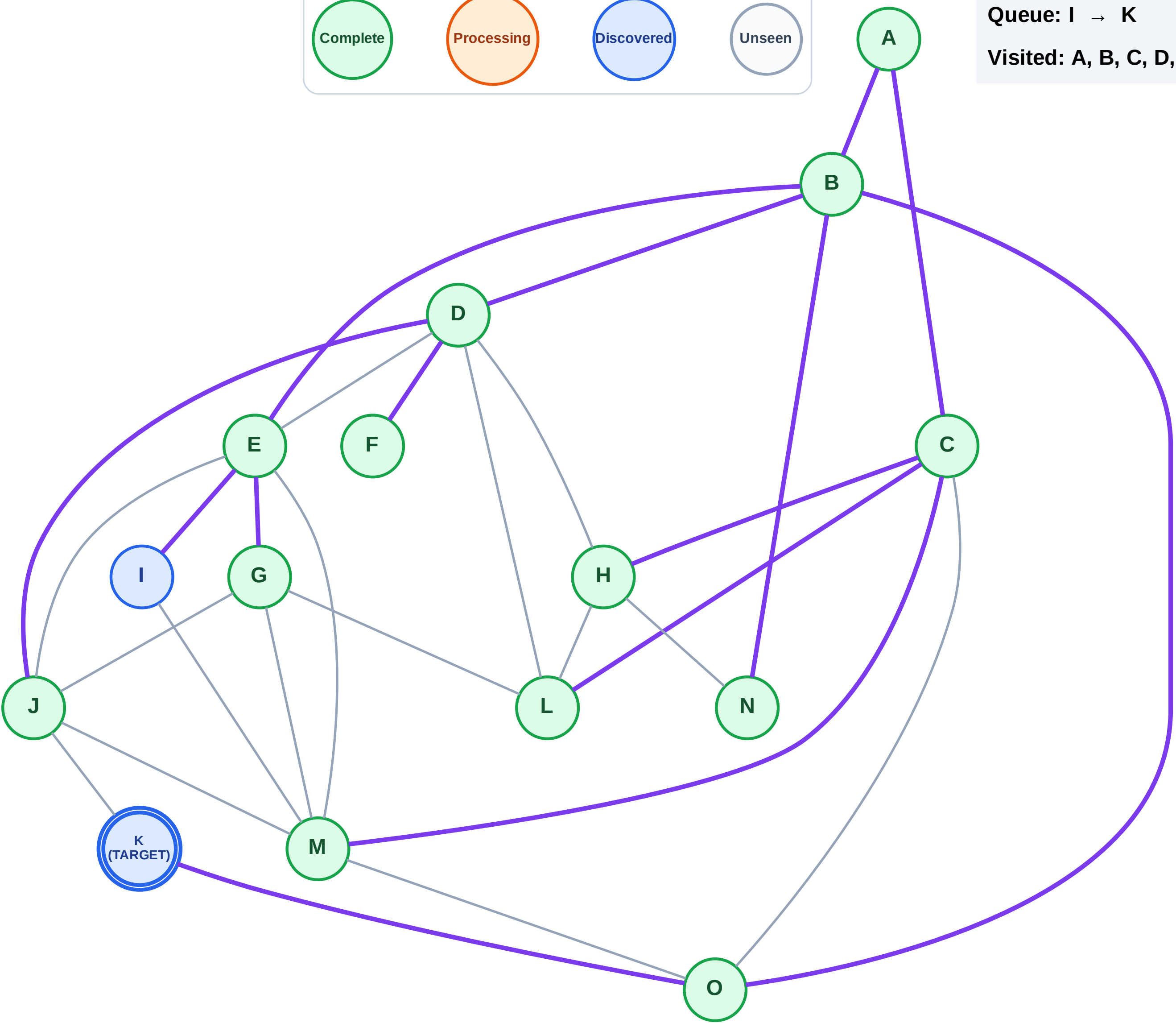
Discovered



Unseen

Queue: I → K

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

Step 28: Process node I

Legend

Complete

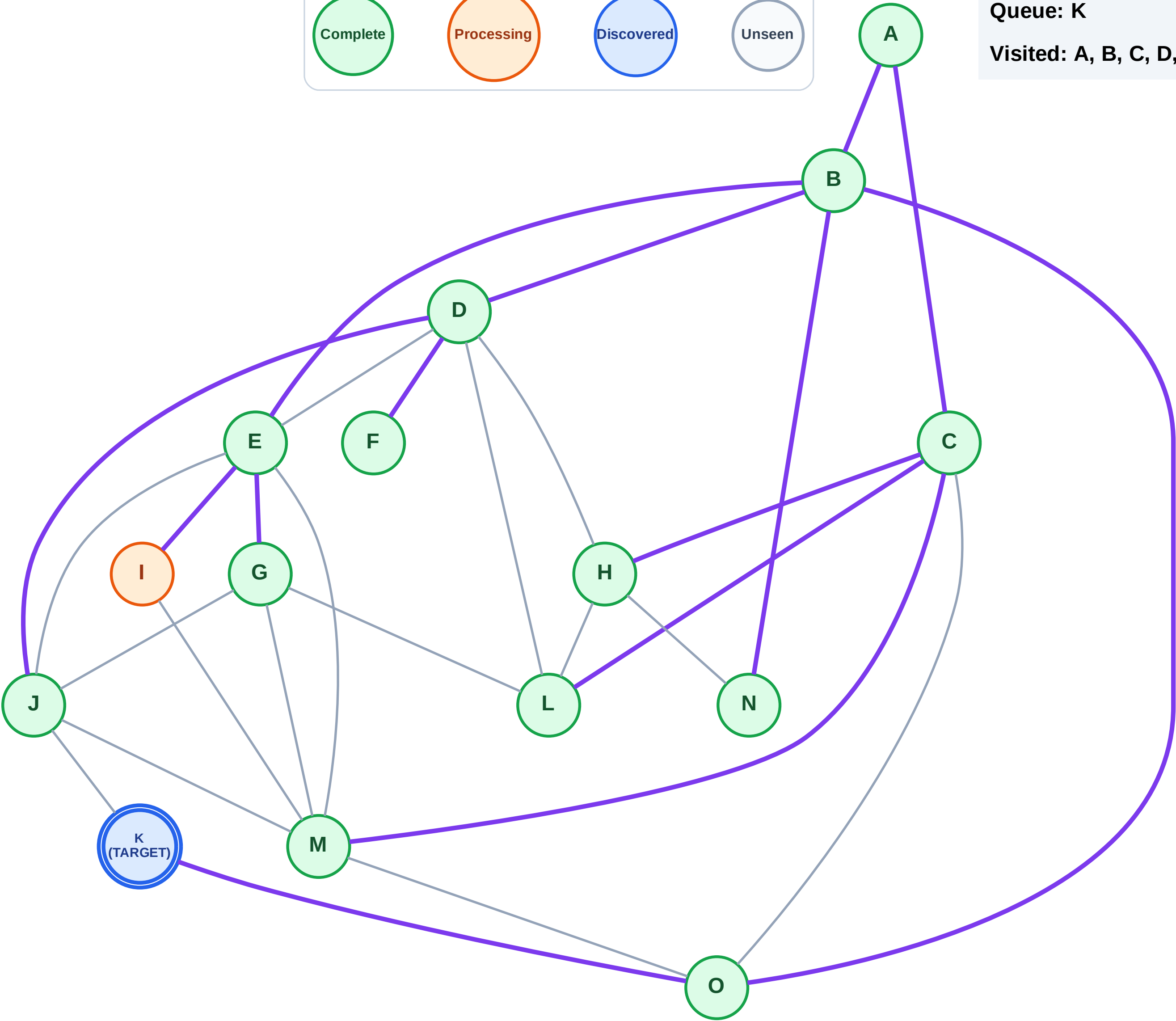
Processing

Discovered

Unseen

Queue: K

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



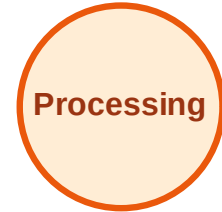
BFS Traversal

Step 29: No new neighbors from I

Legend



Complete



Processing



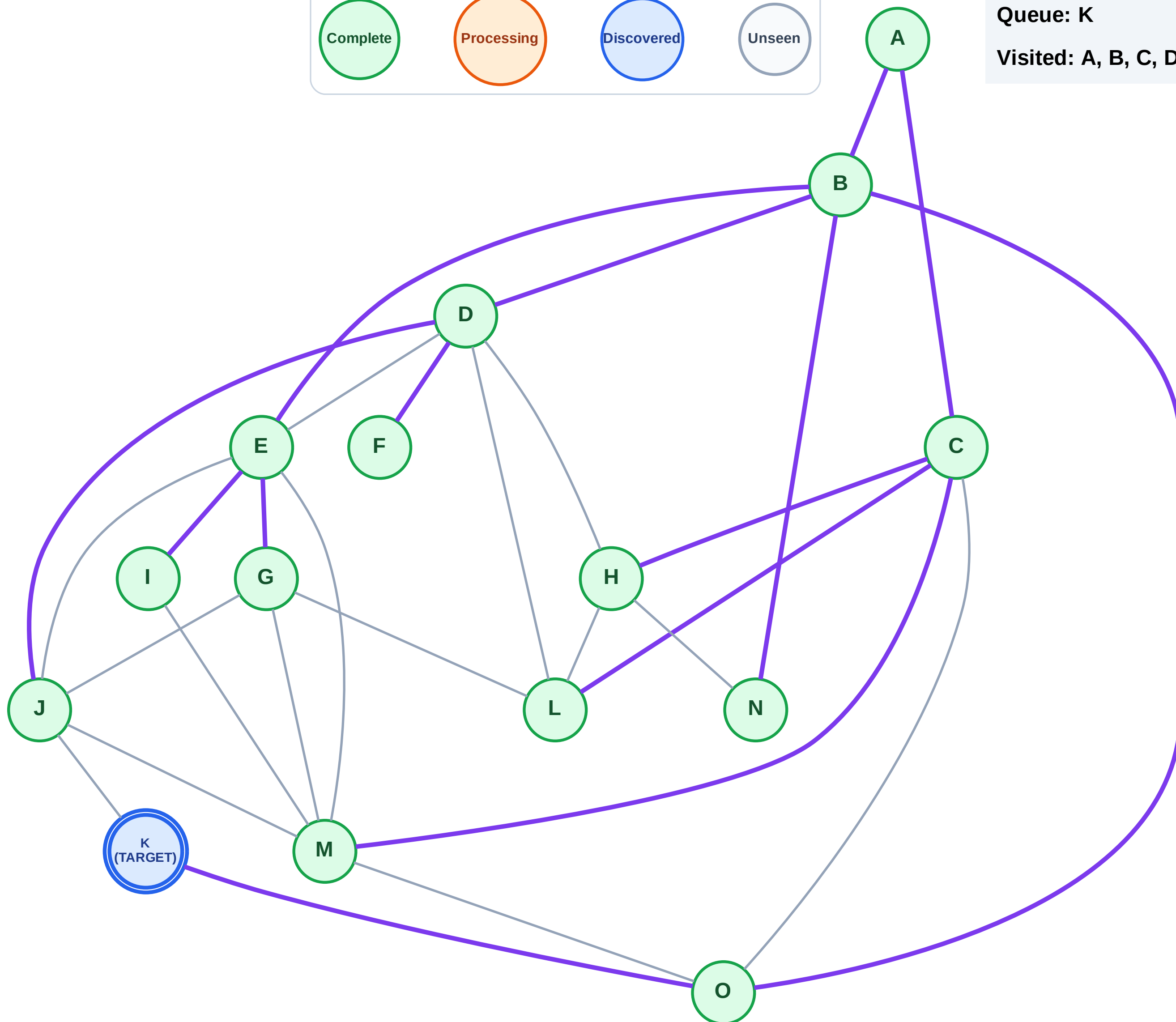
Discovered



Unseen

Queue: K

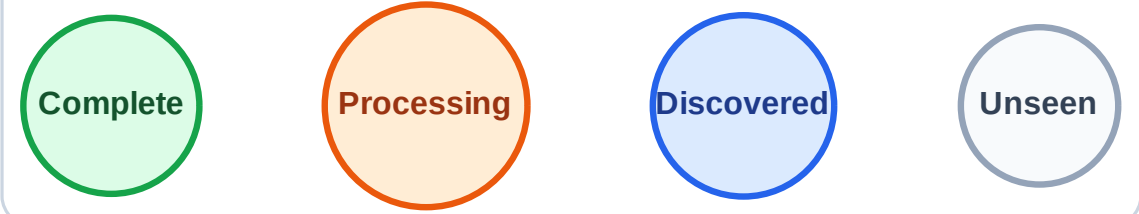
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

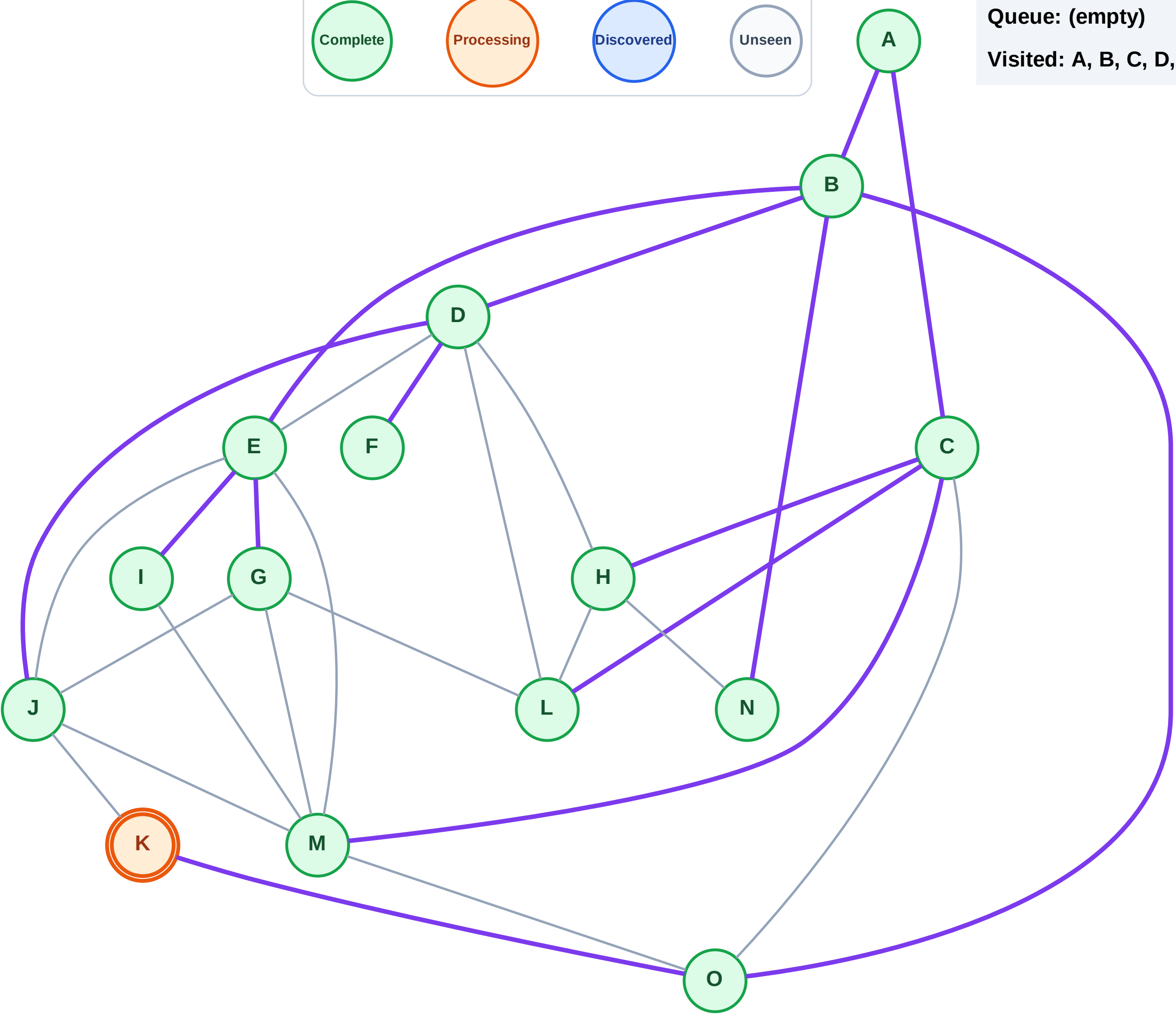
Step 30: Process node K

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 31: TARGET NODE K REACHED! Path (3 hops): A → B → O → K

Legend

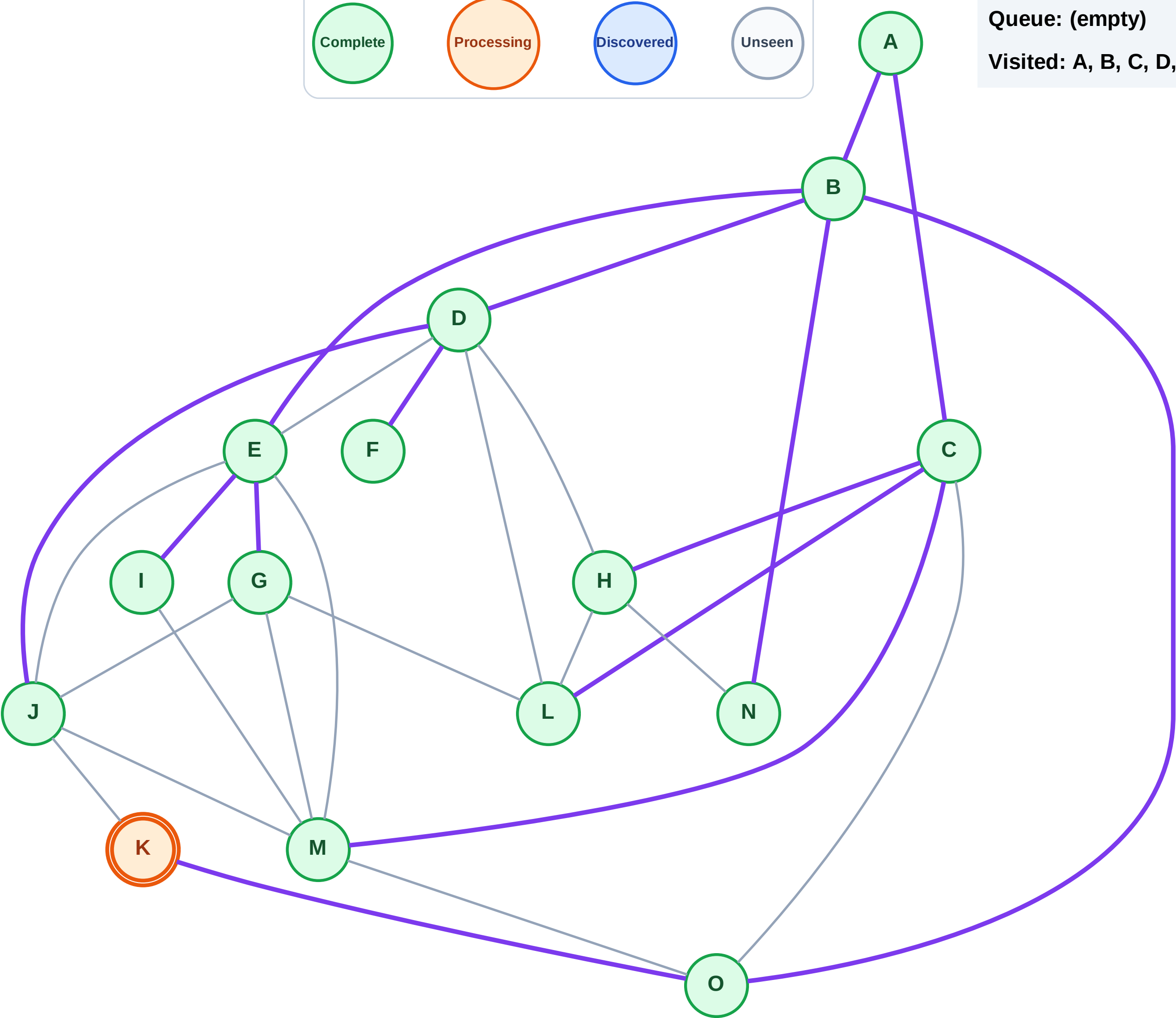
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

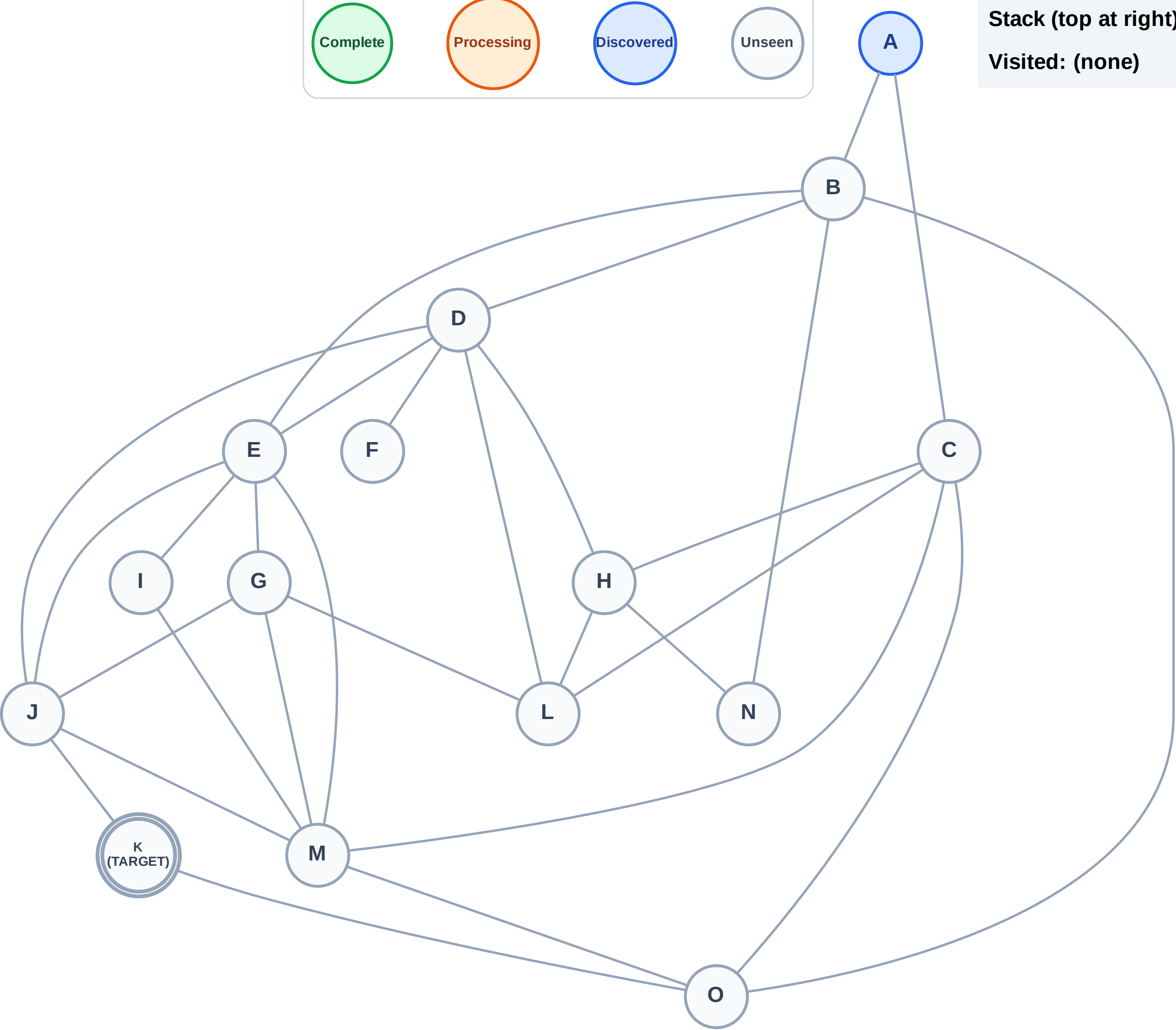
Complete

Processing

Discovered

Unseen

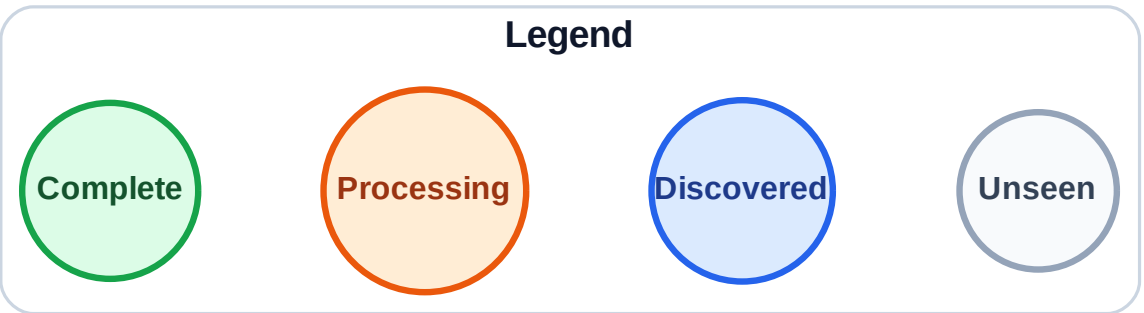
Stack (top at right): A
Visited: (none)



DFS Traversal

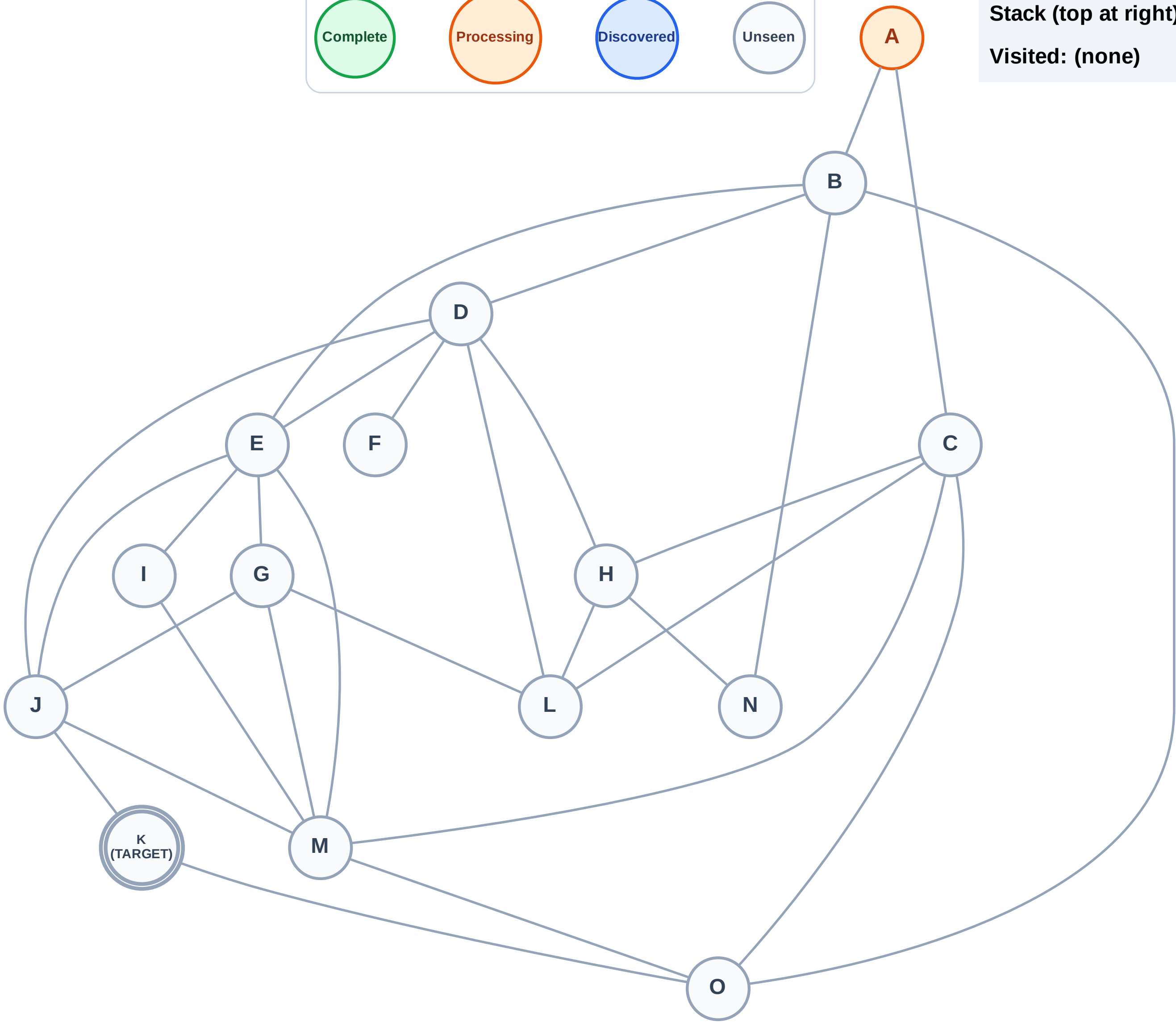
Step 2: Process node A

Legend



Stack (top at right): (empty)

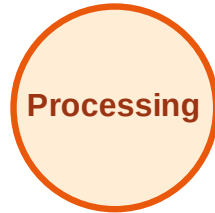
Visited: (none)



DFS Traversal

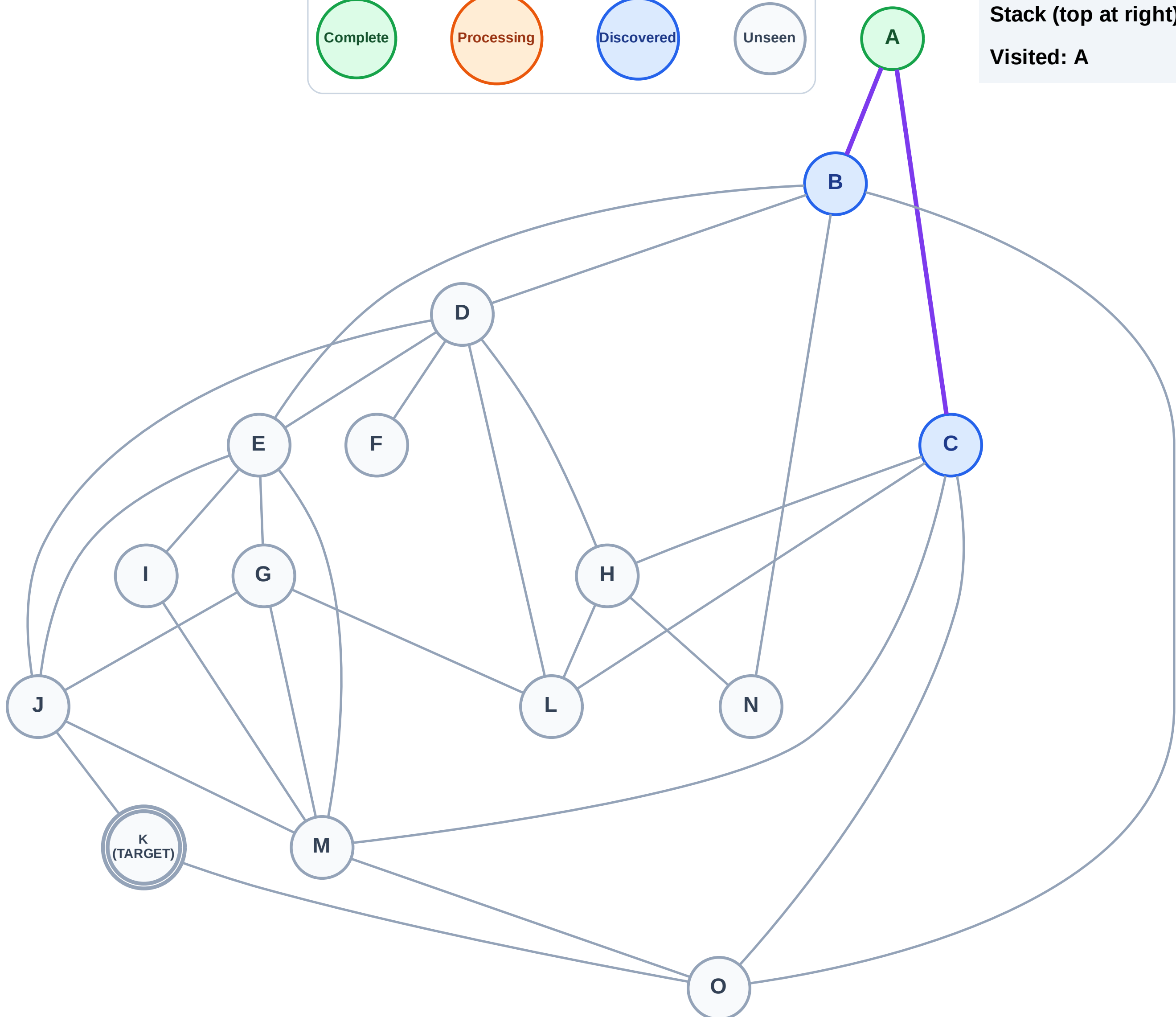
Step 3: Discover C, B from A

Legend



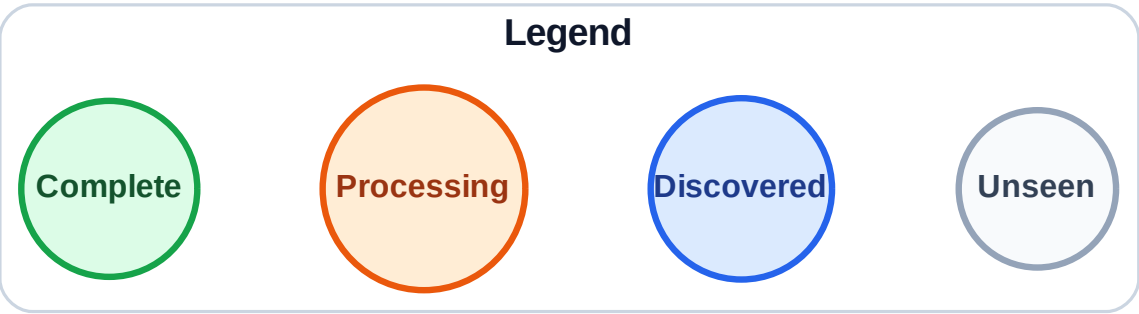
Stack (top at right): C | B

Visited: A



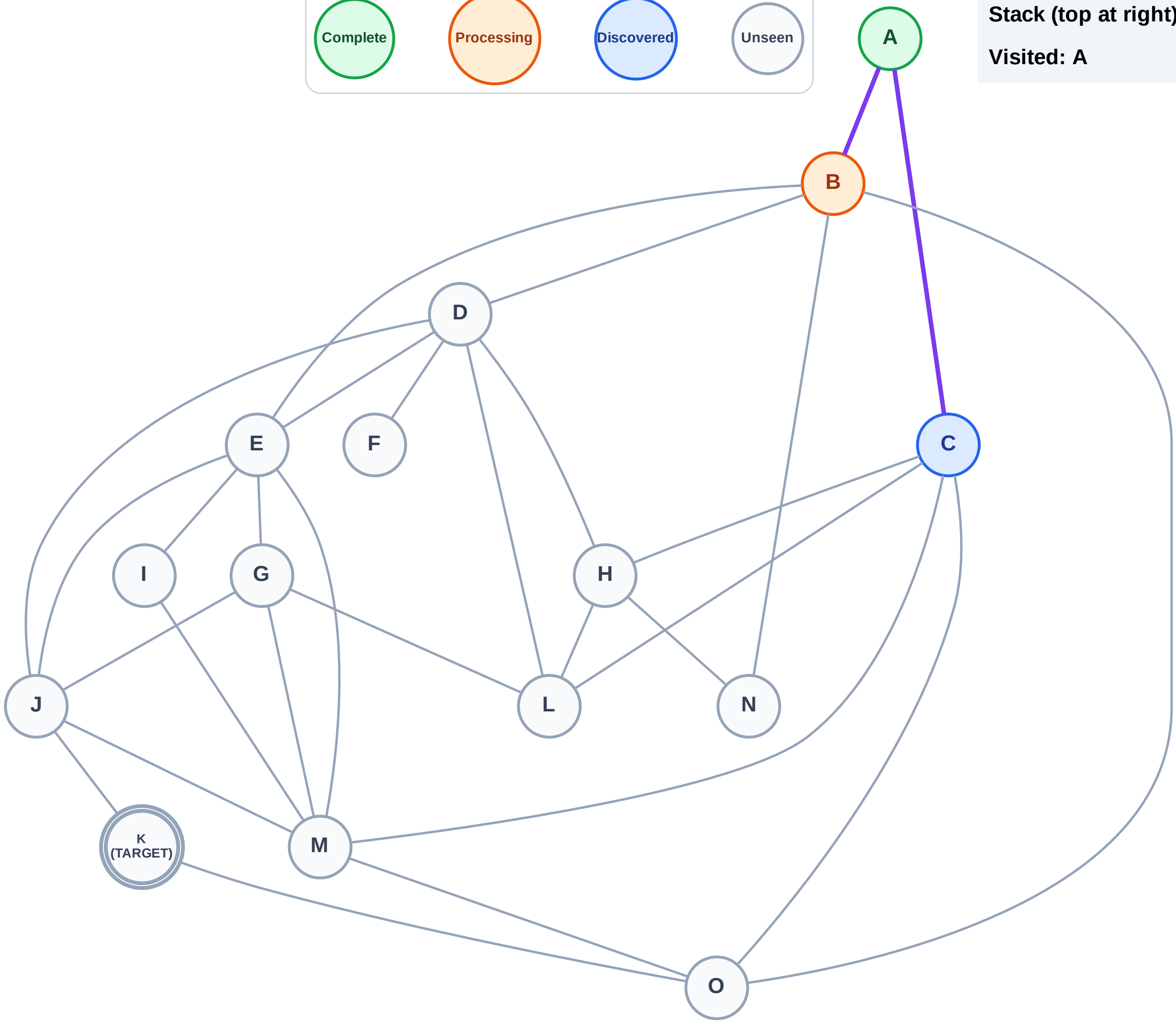
DFS Traversal

Step 4: Process node B



Stack (top at right): C

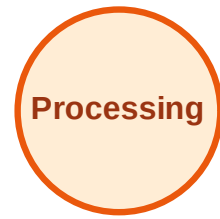
Visited: A



DFS Traversal

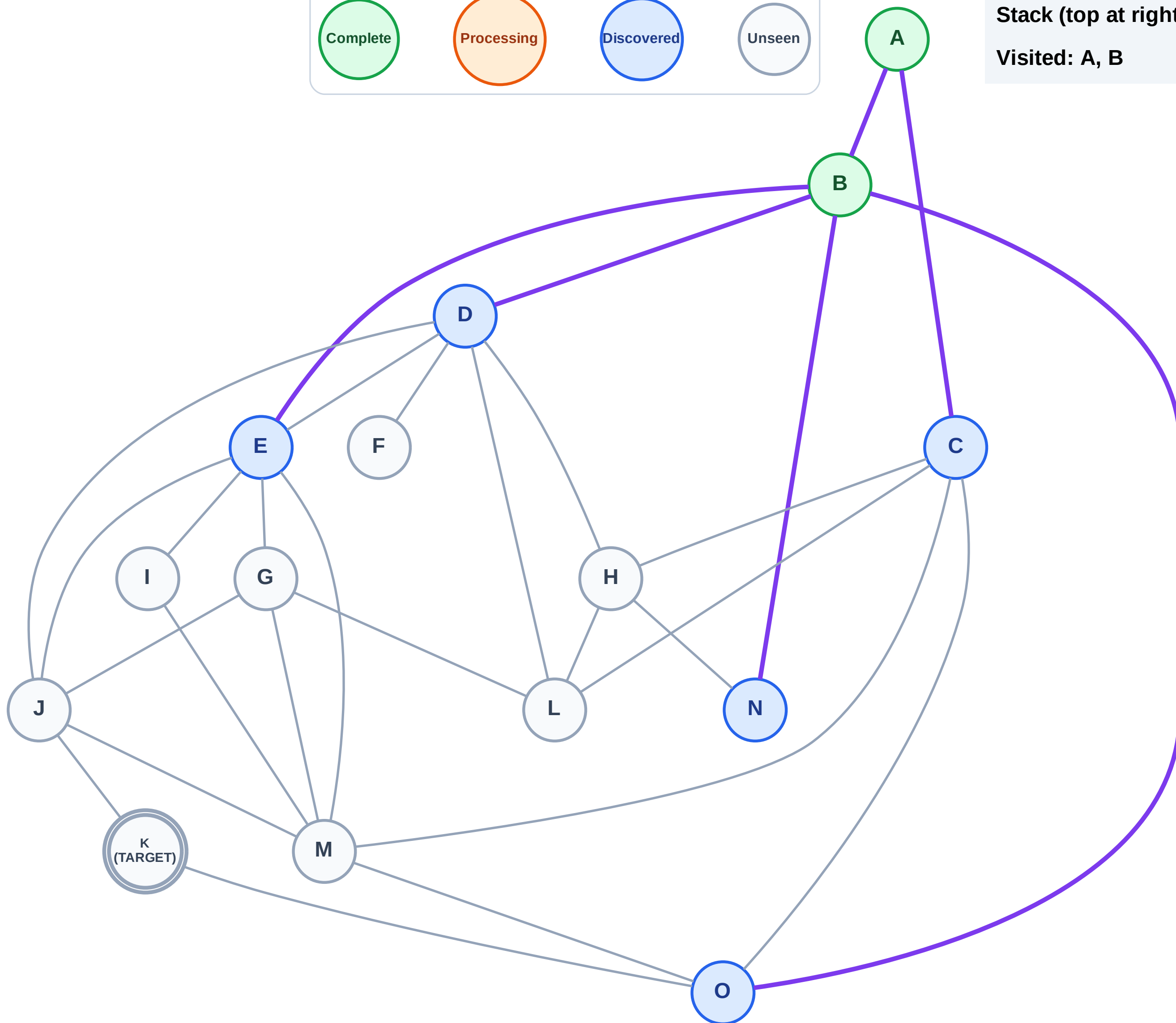
Step 5: Discover O, N, E, D from B

Legend



Stack (top at right): C | O | N | E | D

Visited: A, B



Step 6: Process node D

Step 6: Process node D

Legend

Complete

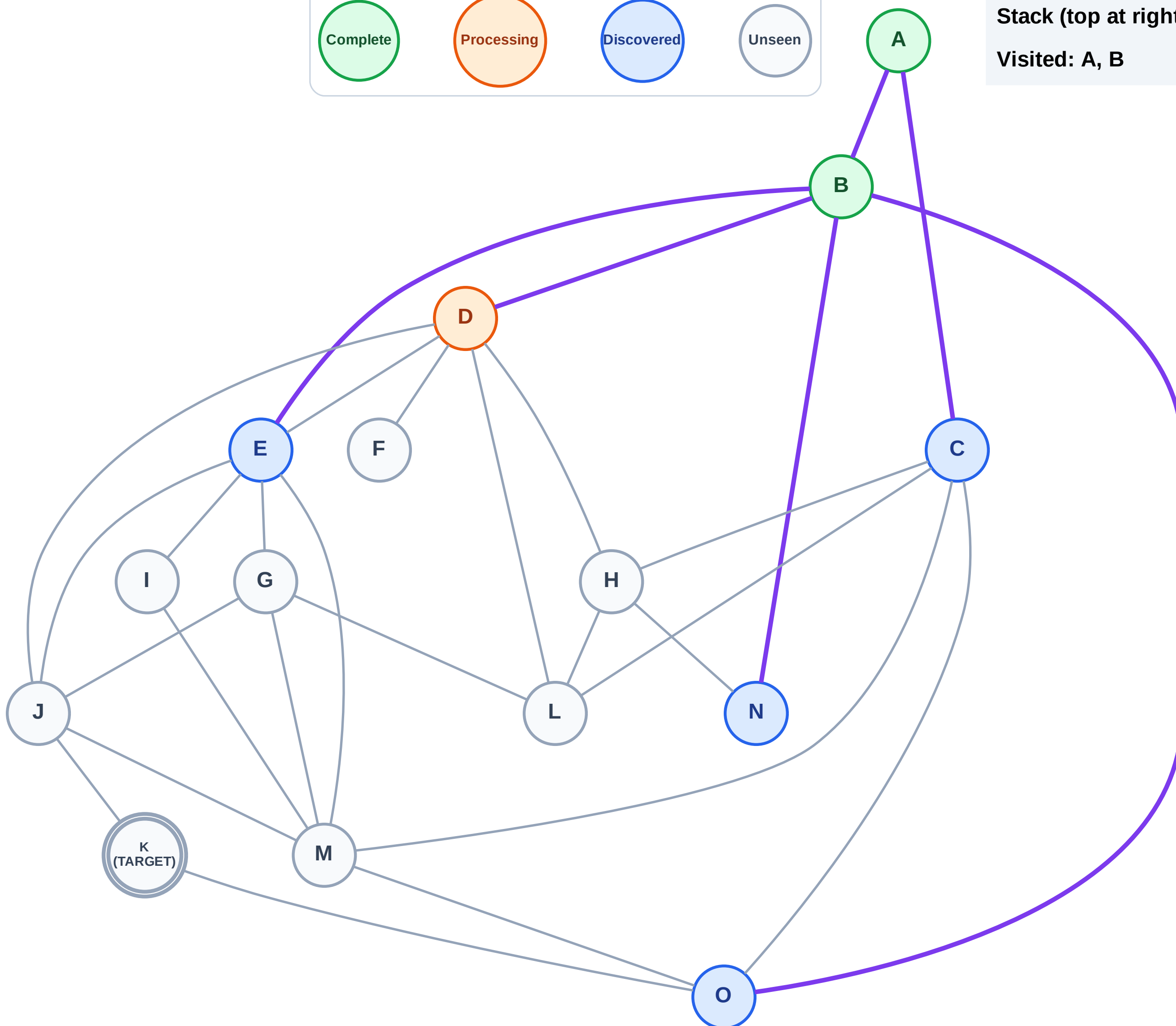
Processing

Discovered

Unseen

Stack (top at right): C | O | N | E

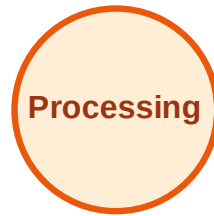
Visited: A, B



DFS Traversal

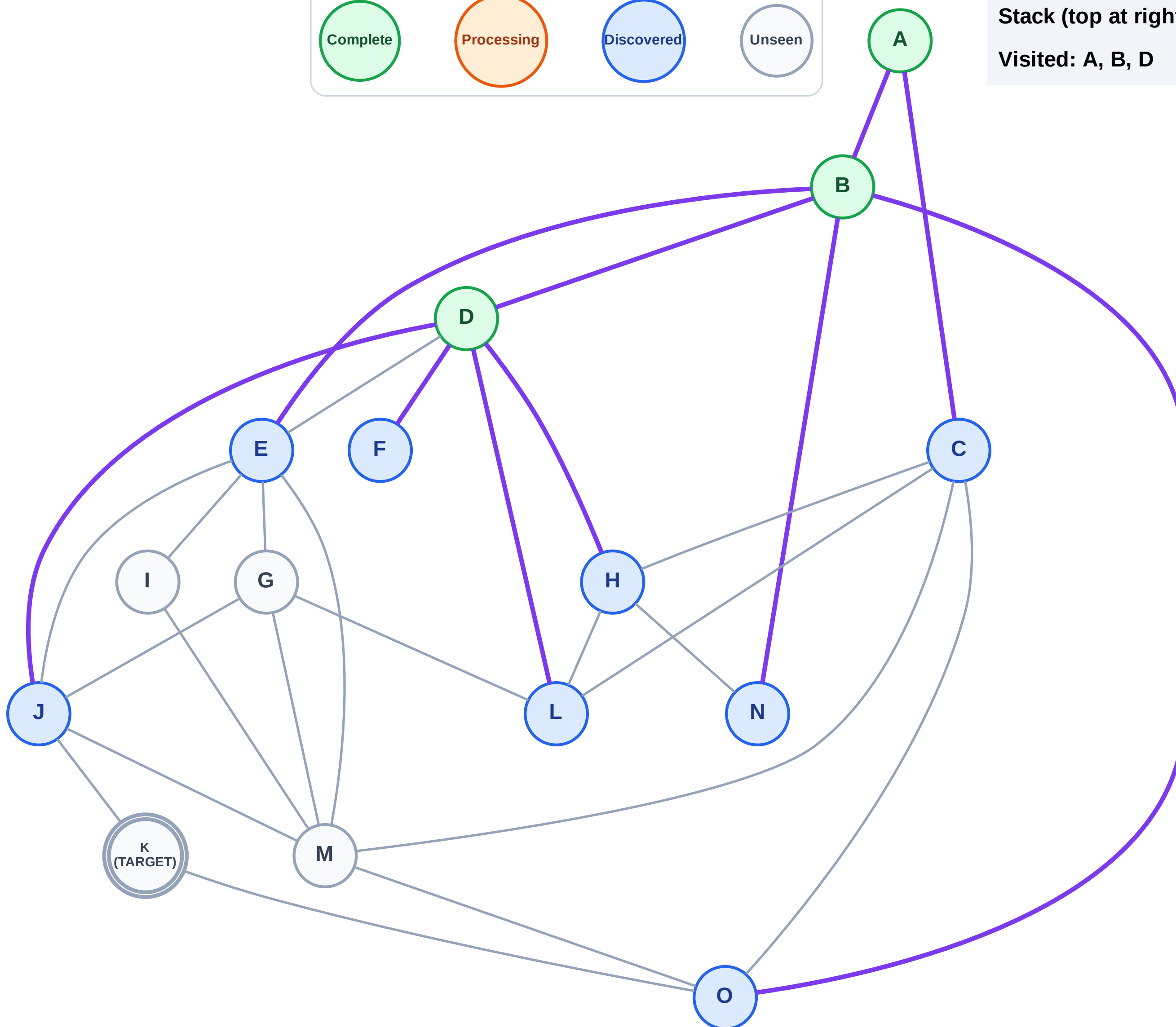
Step 7: Discover L, J, H, F from D

Legend



Stack (top at right): C | O | N | E | L | J | H | F

Visited: A, B, D



DFS Traversal

Step 8: Process node F

Legend

Complete

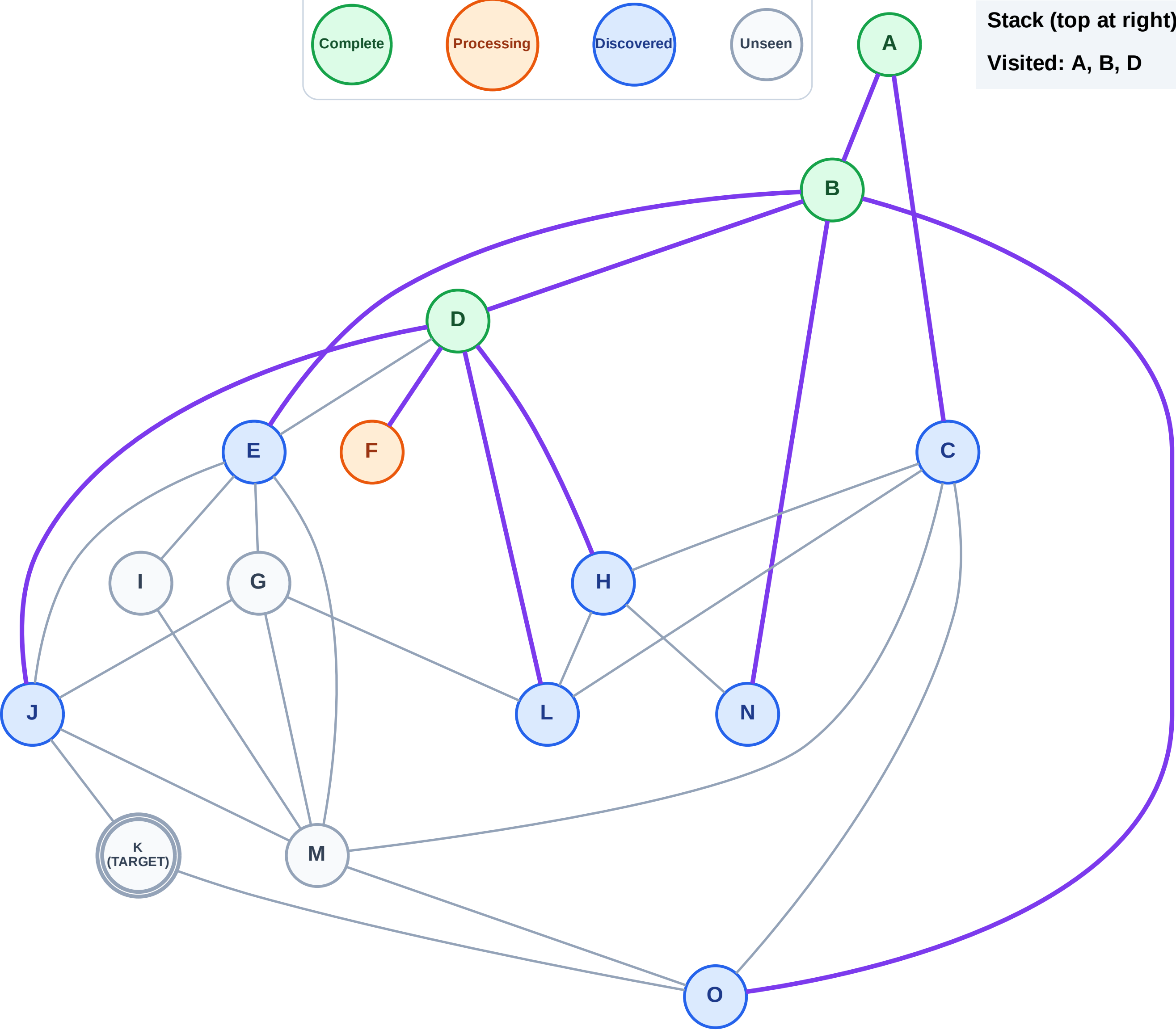
Processing

Discovered

Unseen

Stack (top at right): C | O | N | E | L | J | H

Visited: A, B, D



DFS Traversal

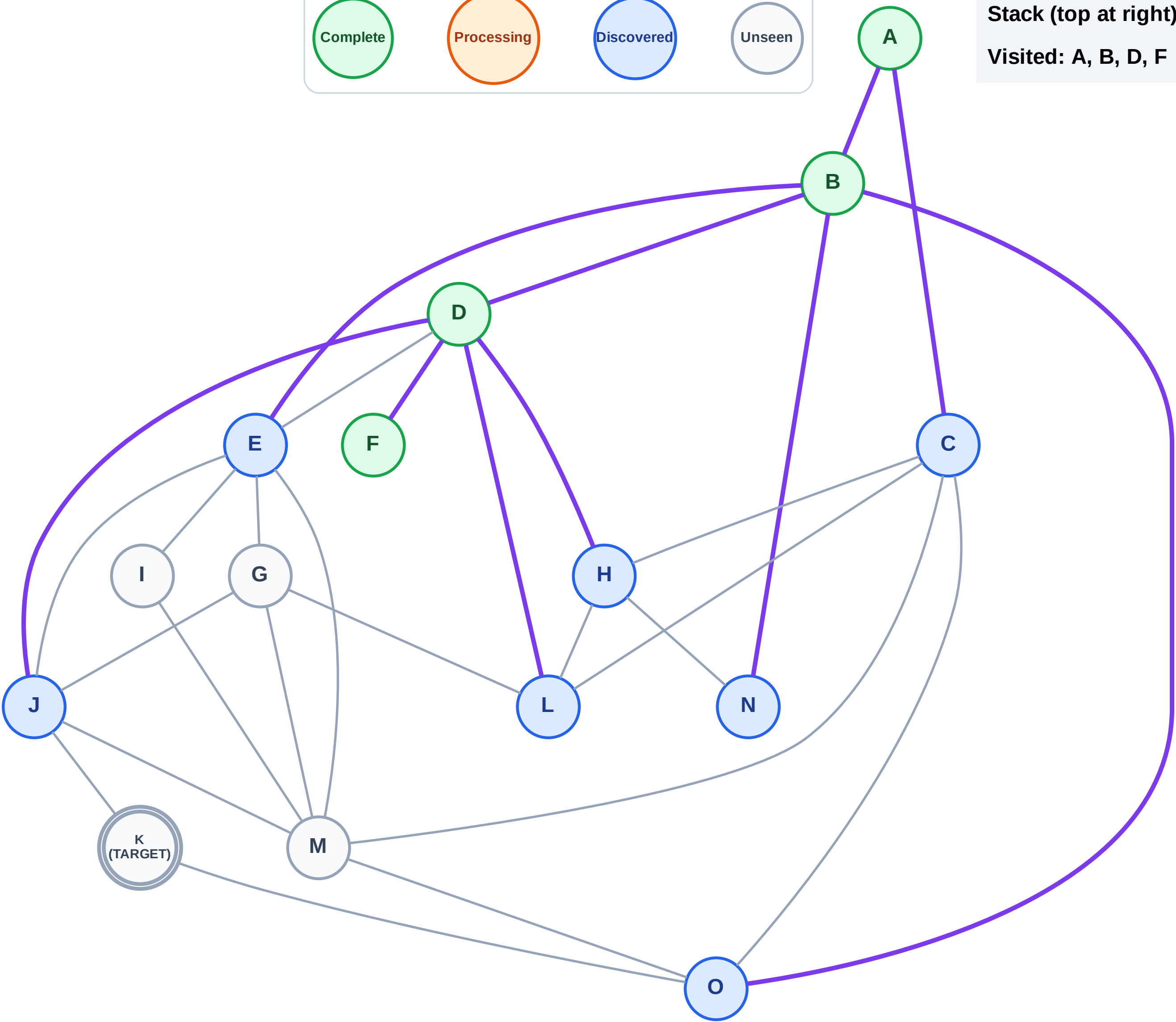
Step 9: No new neighbors from F

Legend



Stack (top at right): C | O | N | E | L | J | H

Visited: A, B, D, F



DFS Traversal

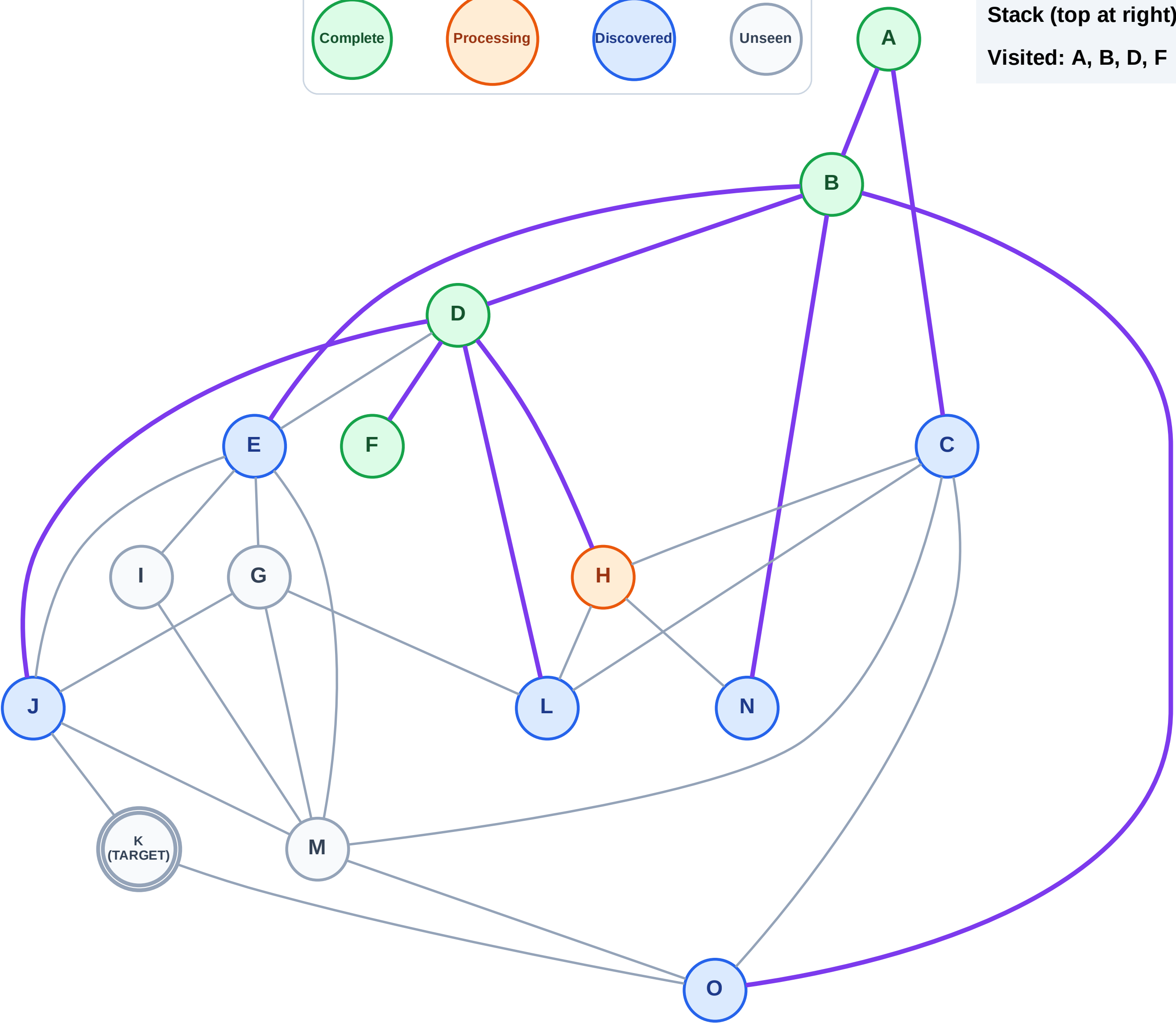
Step 10: Process node H

Legend



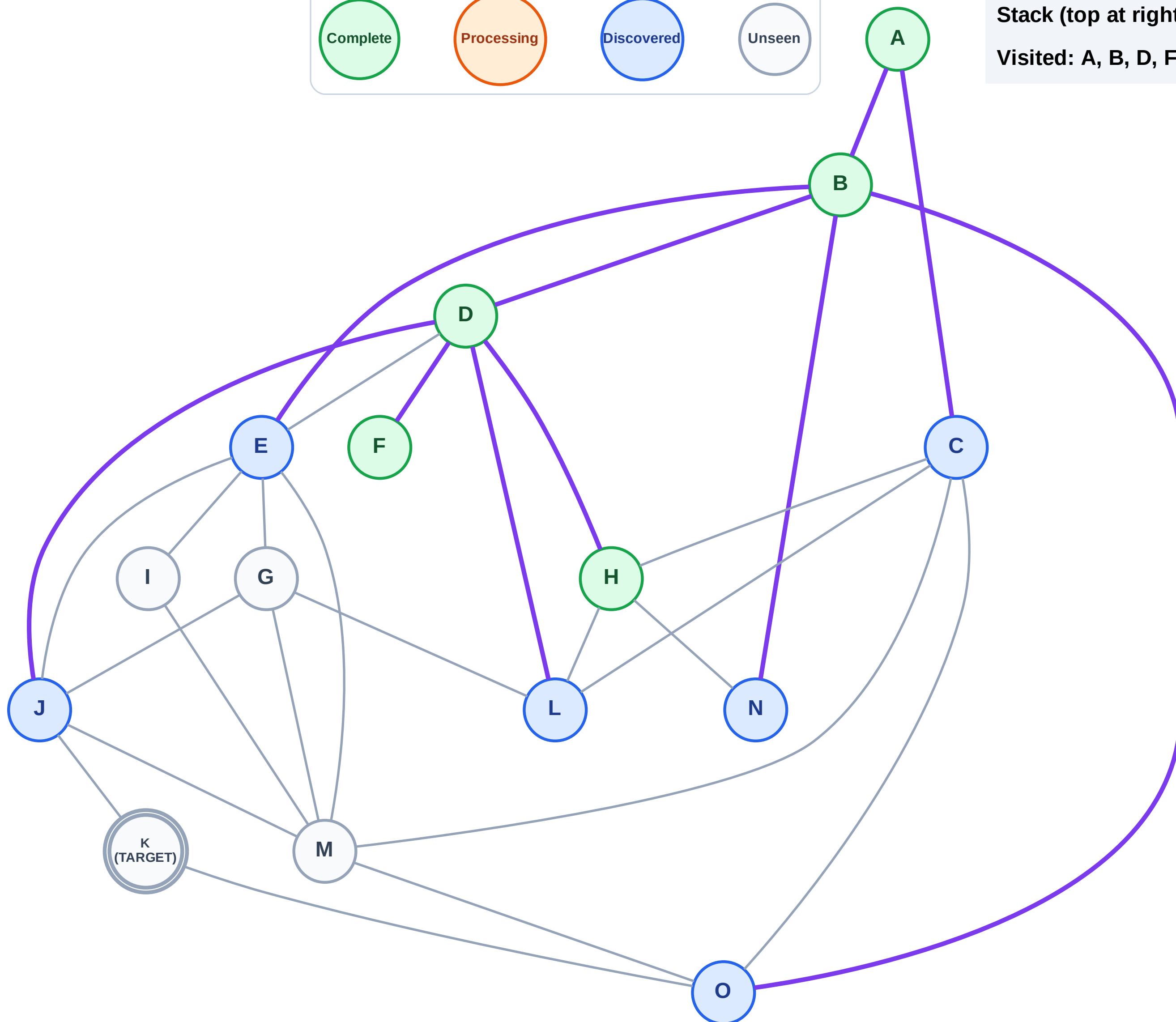
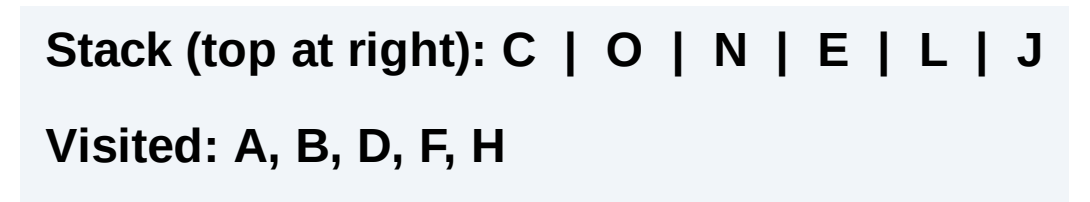
Stack (top at right): C | O | N | E | L | J

Visited: A, B, D, F



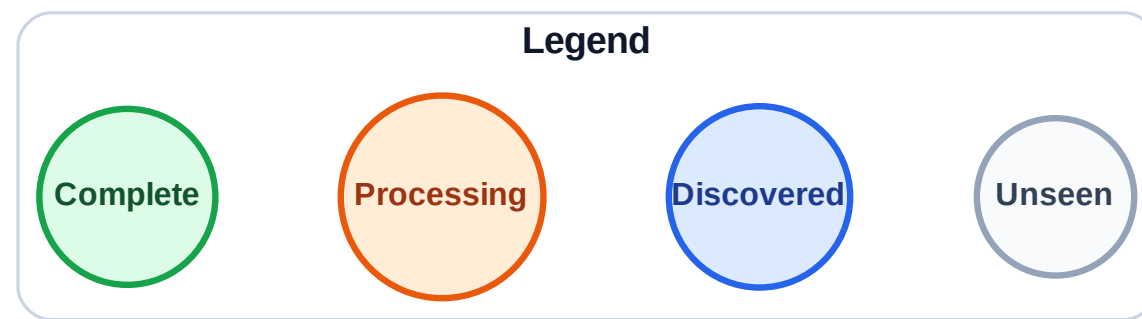
Step 11: No new neighbors from H

Step 11: No new neighbors from H



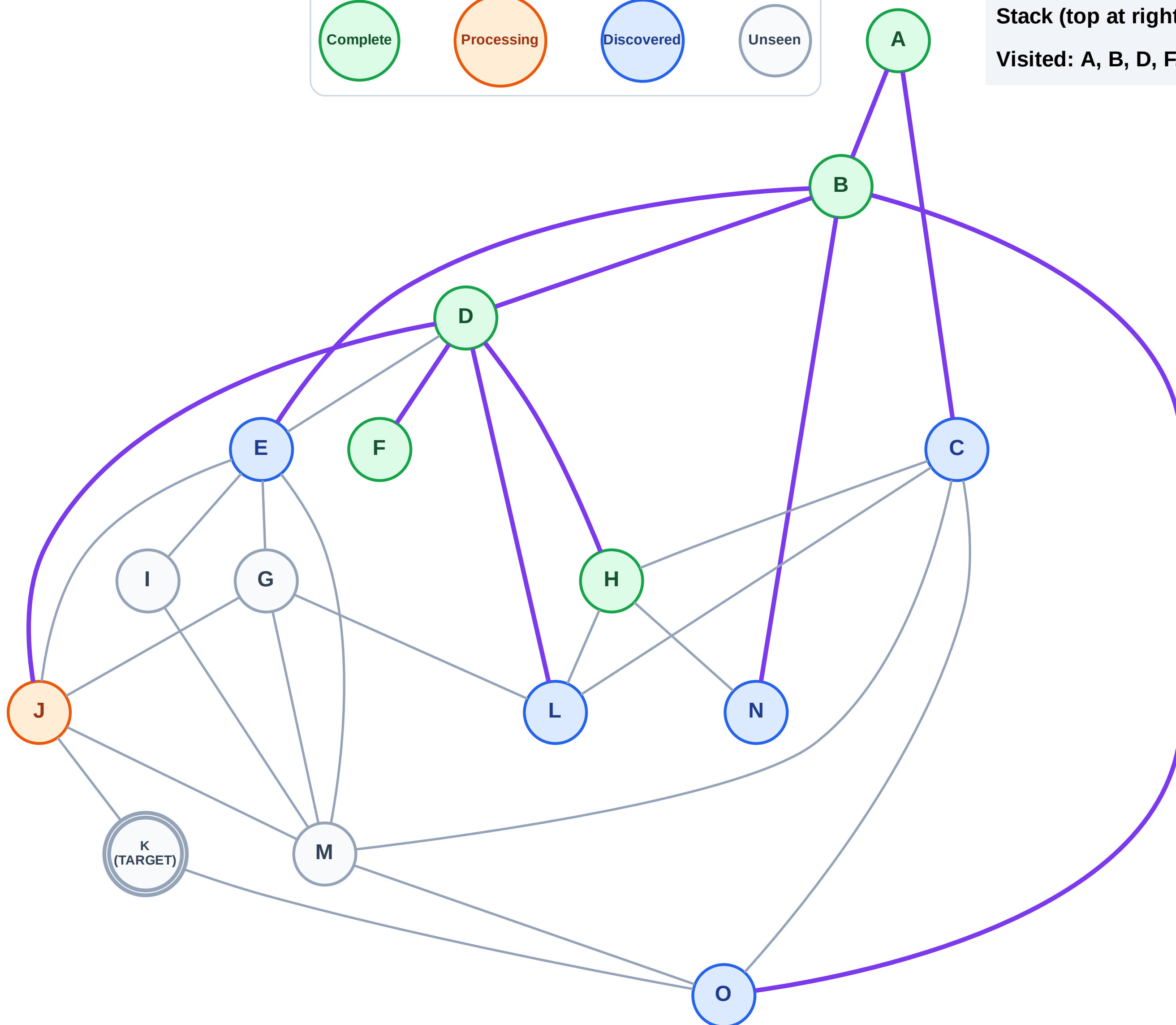
DFS Traversal

Step 12: Process node J



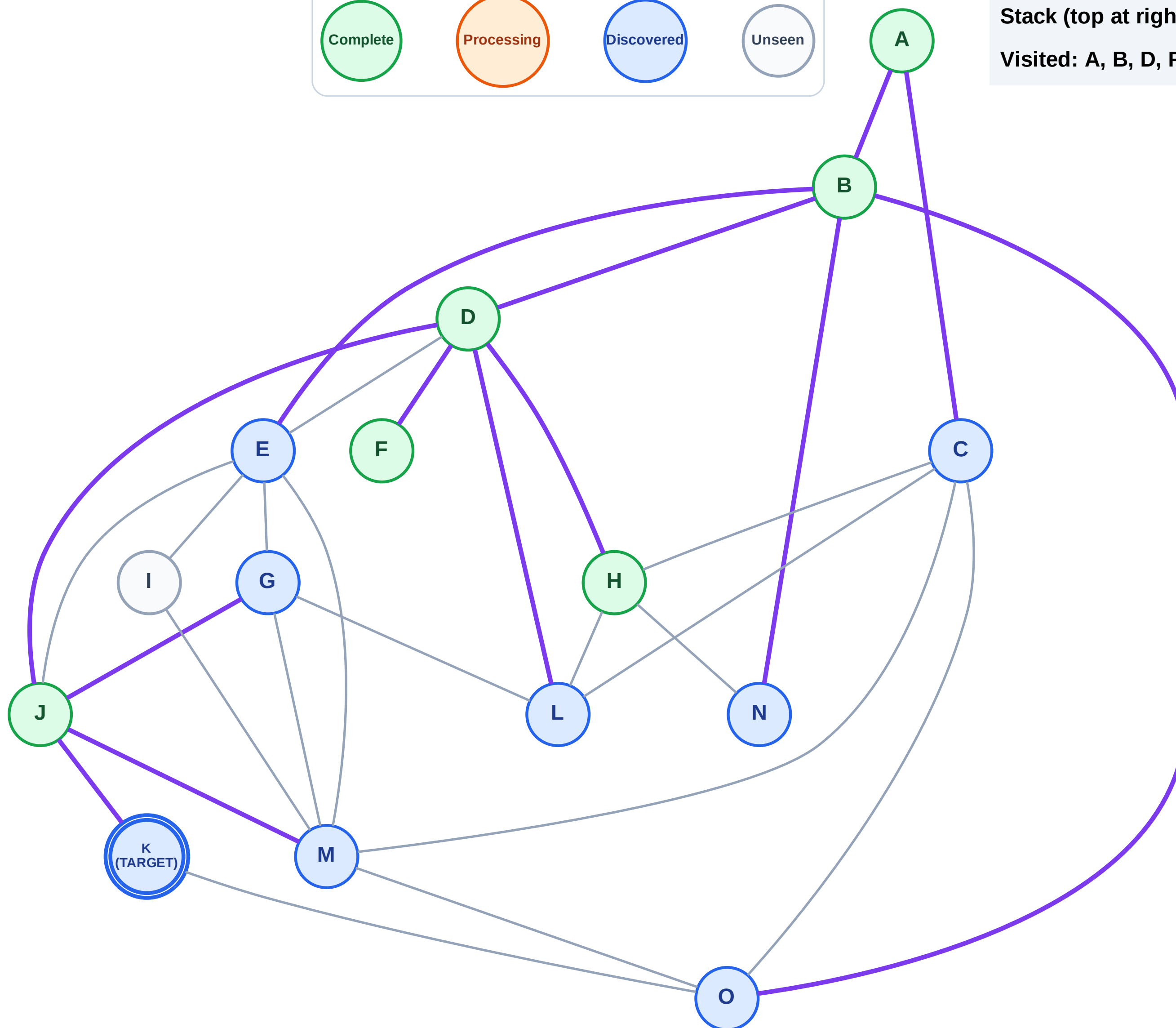
Stack (top at right): C | O | N | E | L

Visited: A, B, D, F, H



Step 13: Discover M, K, G from J

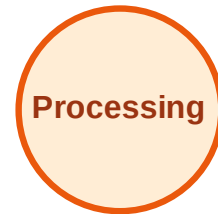
Step 13: Discover M, K, G from J



DFS Traversal

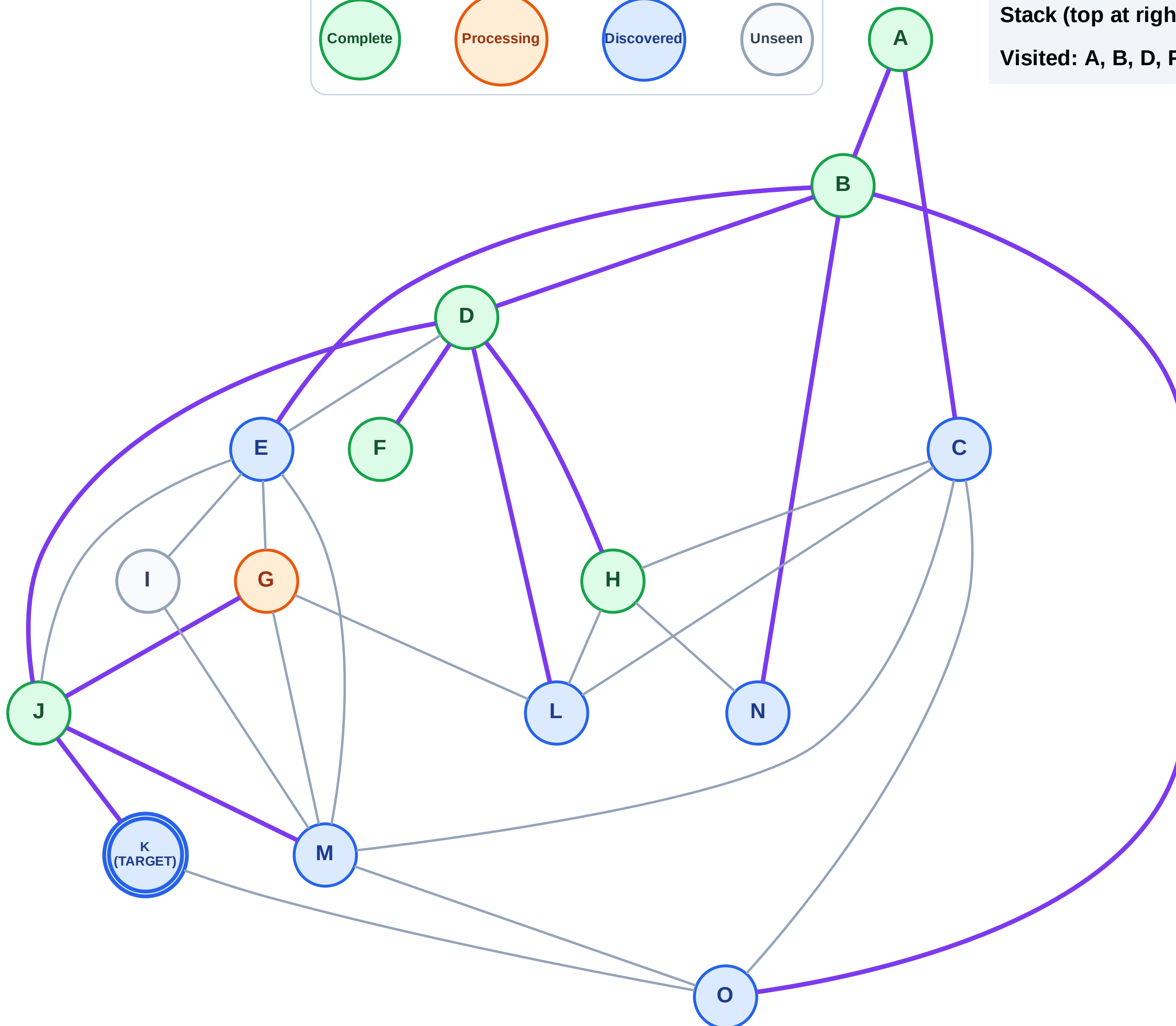
Step 14: Process node G

Legend



Stack (top at right): C | O | N | E | L | M | K

Visited: A, B, D, F, H, J

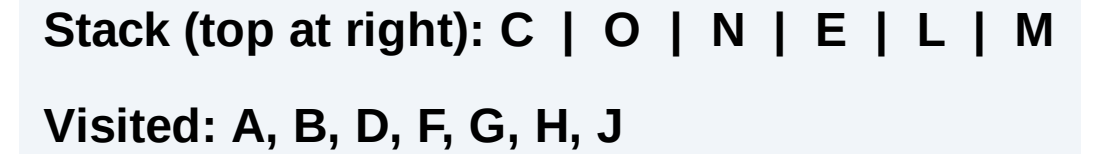


Step 15: No new neighbors from G

Step 15: No new neighbors from G



Step 16: Process node K



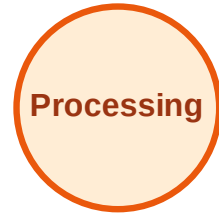
DFS Traversal

Step 17: TARGET NODE K REACHED! Path (4 hops): A → B → D → J → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): C | O | N | E | L | M

Visited: A, B, D, F, G, H, J, K

